



Species Interactions

Continuous-Time Animal Movement Modeling

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Overview

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- 1) Introduction
- 2) Pairwise distances
- 3) A proximity statistic
- 4) Home-range overlap
- 5) Encounter probability & Encounter rates
- 6) Conditional distribution of encounters
- 7) Future directions

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Encounters and Interactions

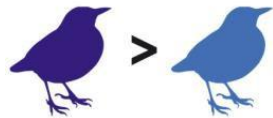
How can we relate individual behavior with higher level processes?

- Studying animal interactions through encounter theory could inform us about:
 - Sociality, life history, predator-prey dynamics, disease transmission, etc.

Higher resolution

(5 s intervals)

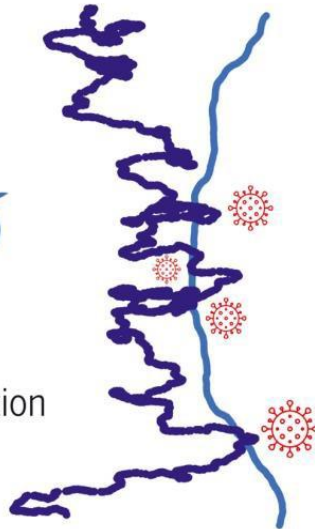
Exploration



Bold

Shy

Multiple interaction hotspots



Individual variation:

Are shy birds less explorative than bold ones?

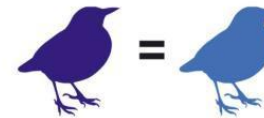
Biotic interactions:

What is the potential for disease transmission?

Lower resolution

(30 min intervals)

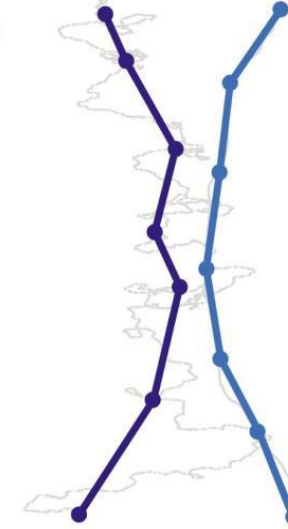
Exploration



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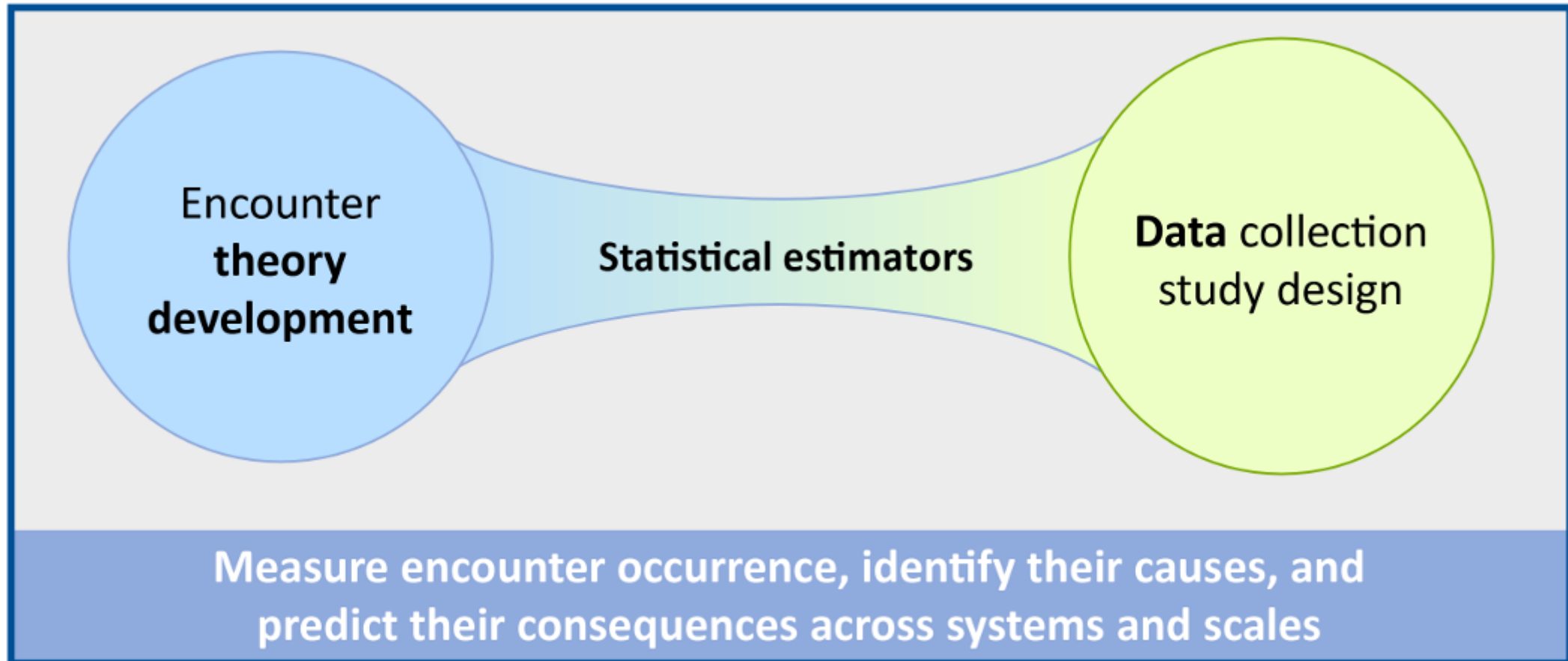
Shy

No interactions



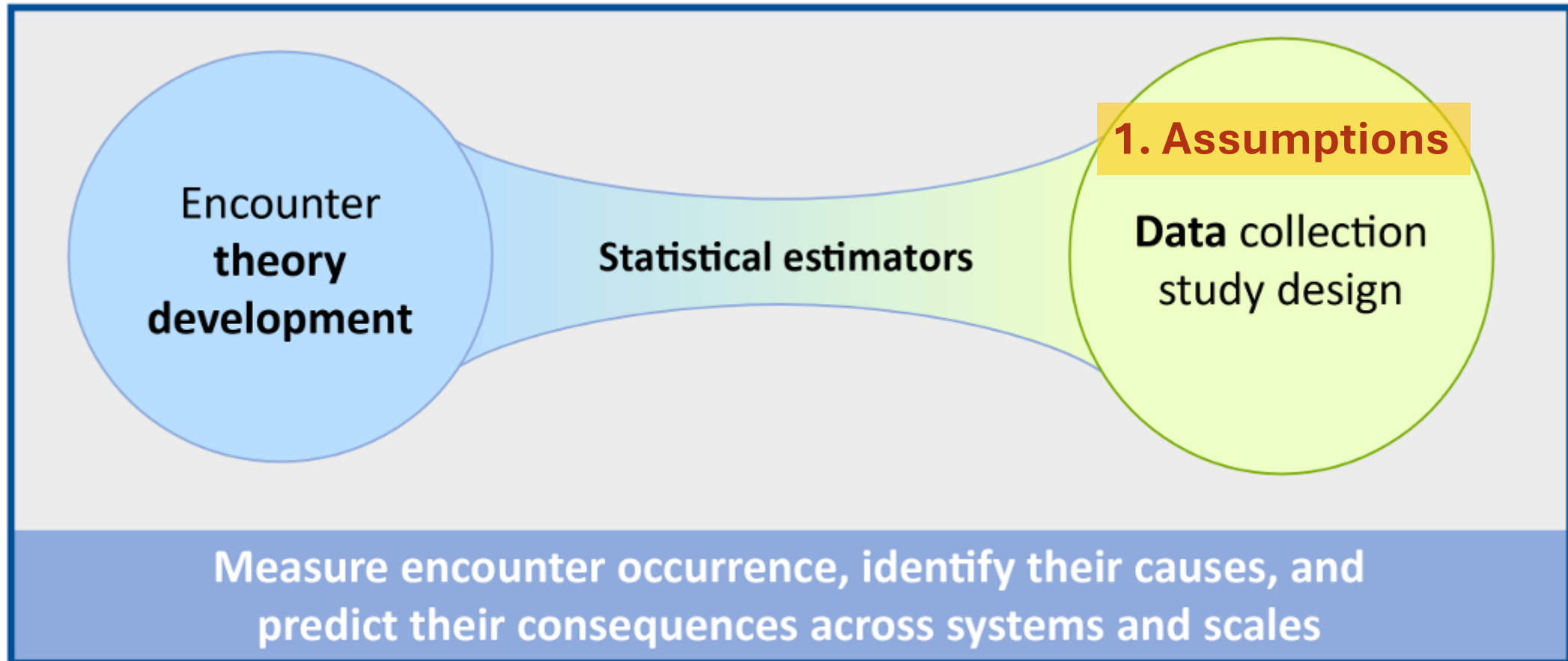
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Towards a general framework for **encounter problems**:



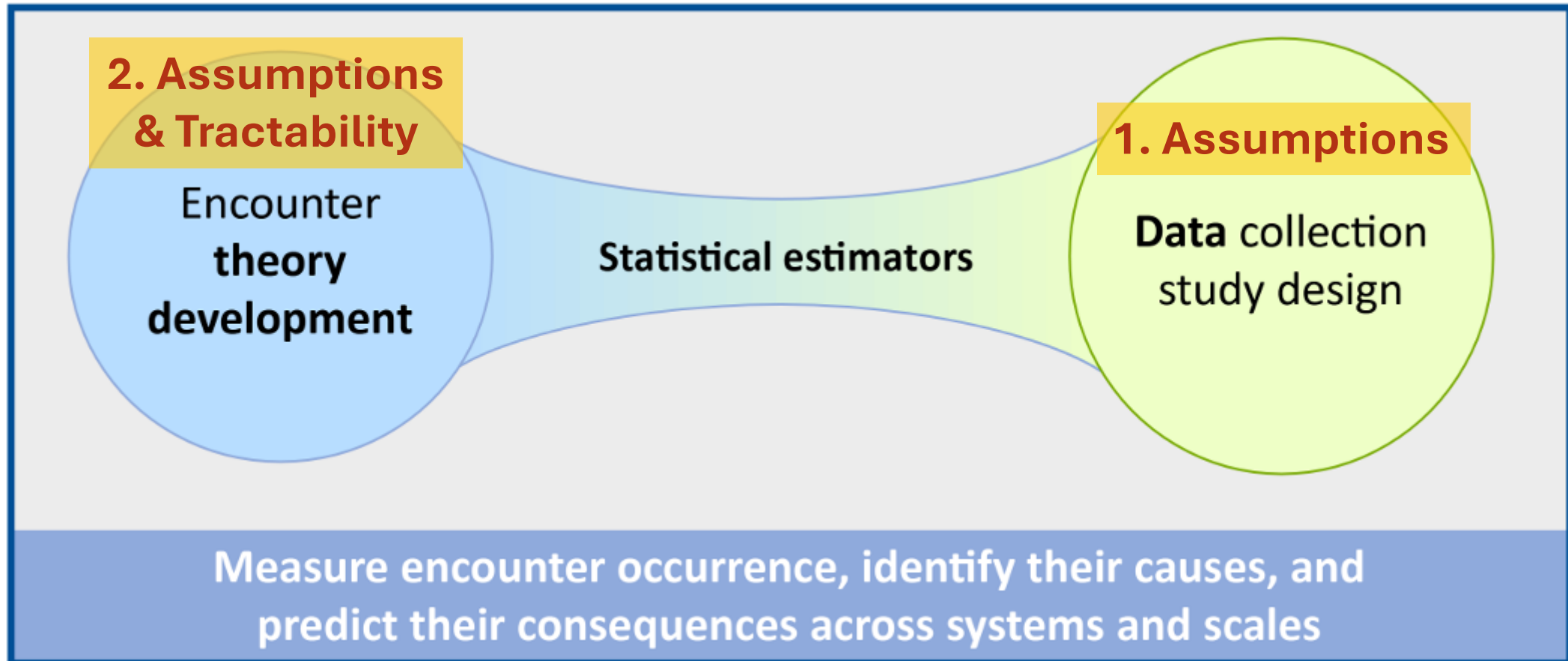
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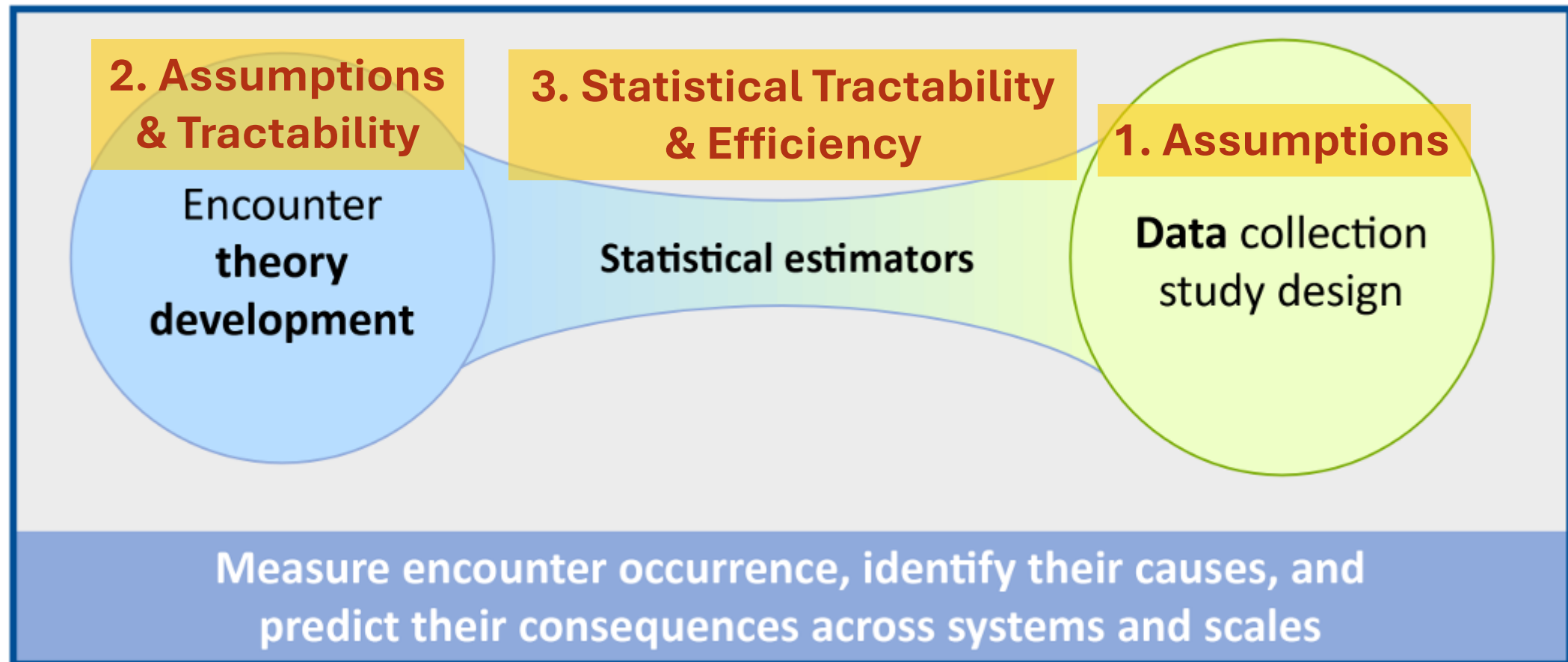
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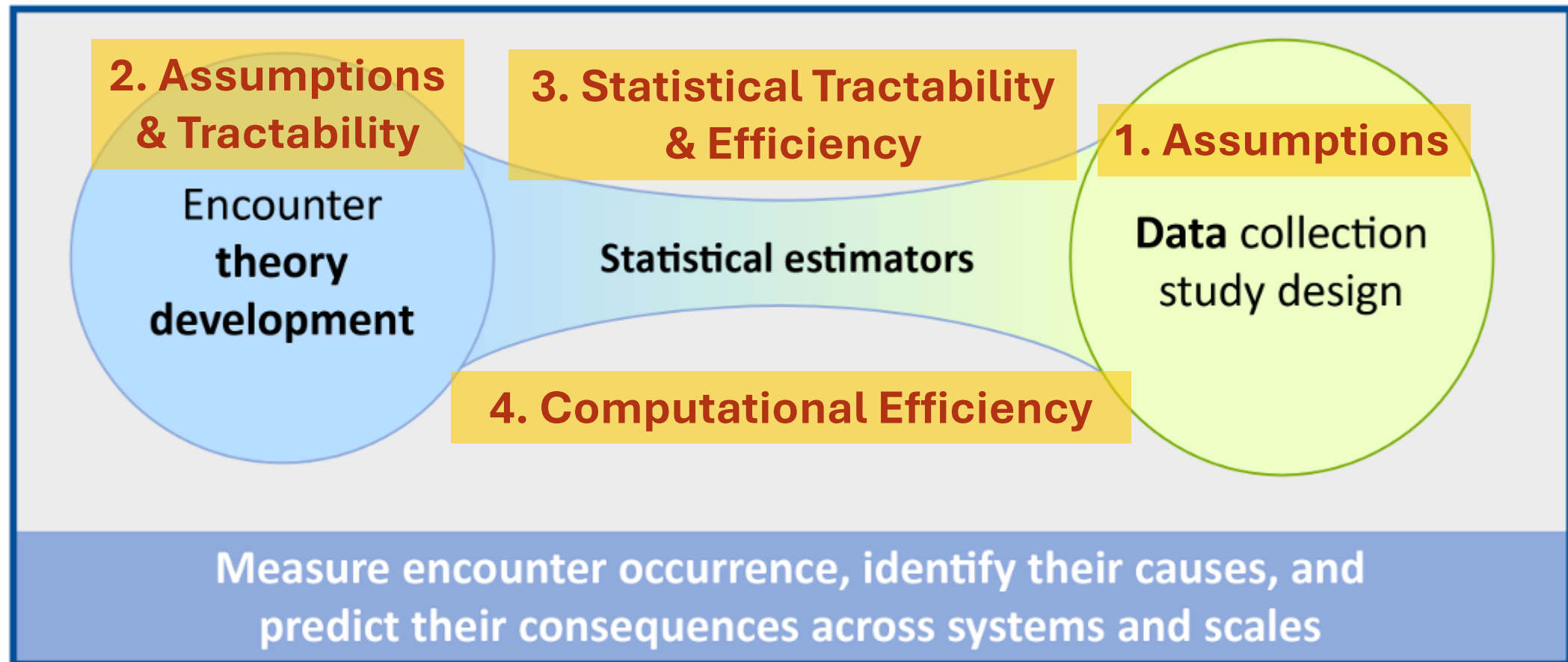
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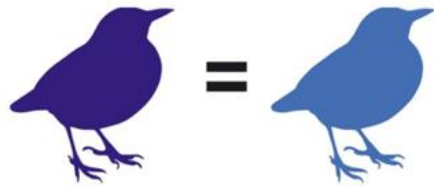


Encounter estimation & prediction

Lower resolution

(30 min intervals)

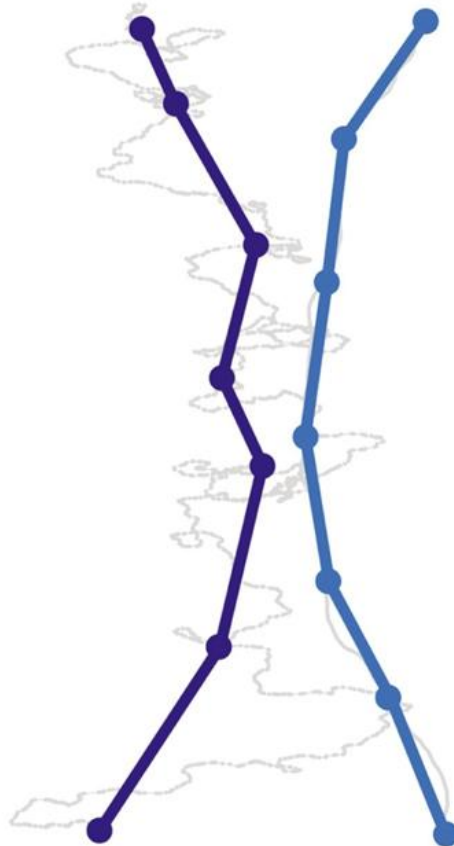
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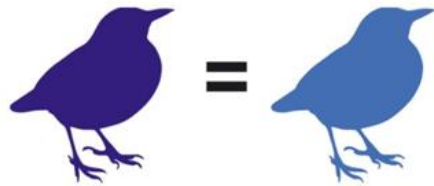


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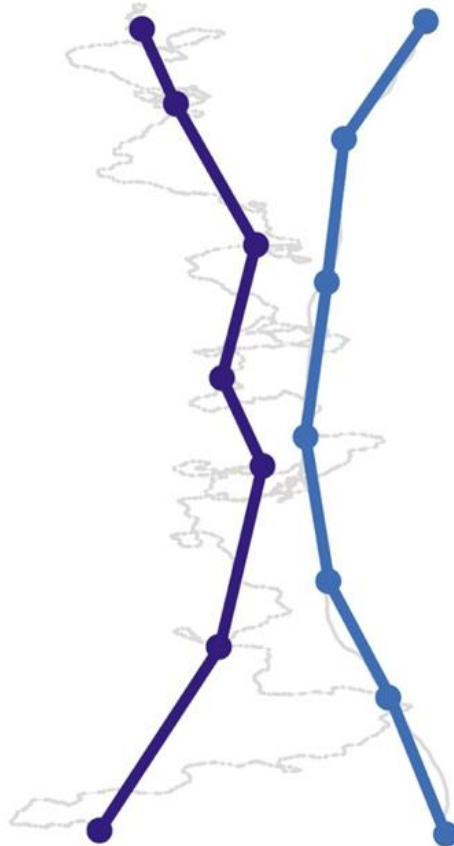
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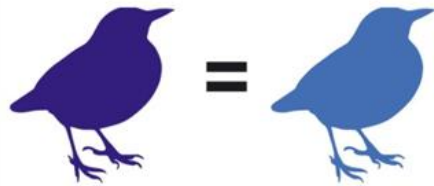
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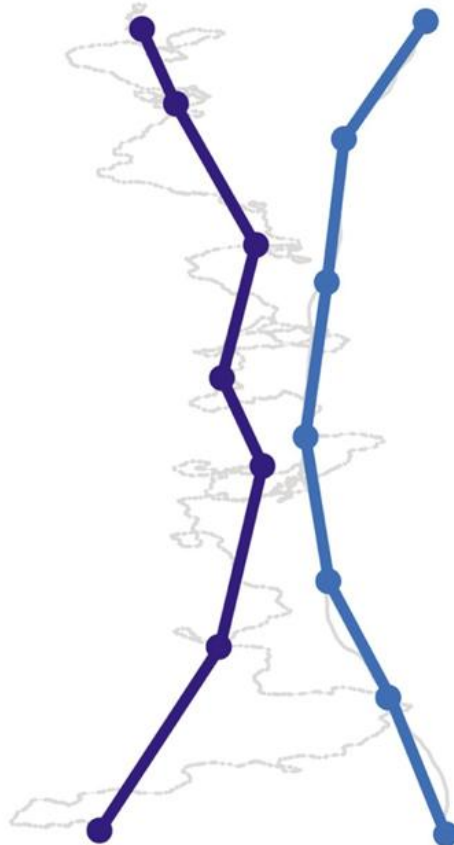
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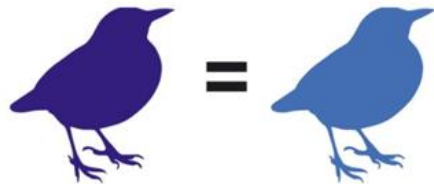
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Encounter estimation & prediction

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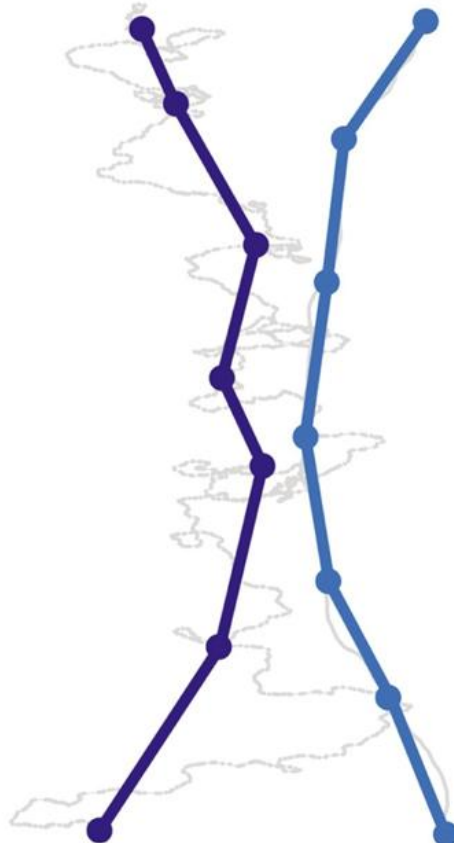
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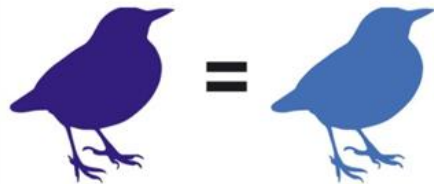
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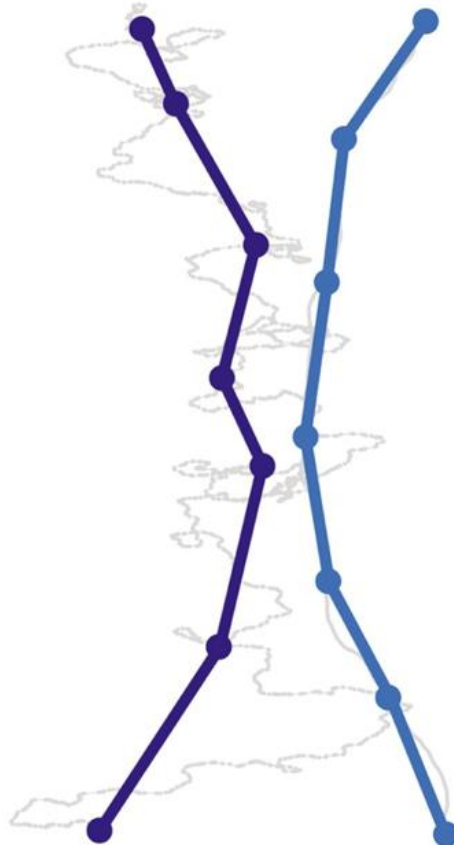
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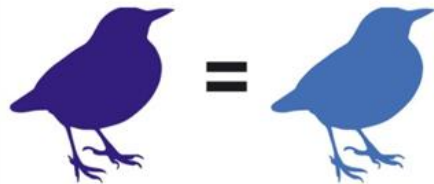
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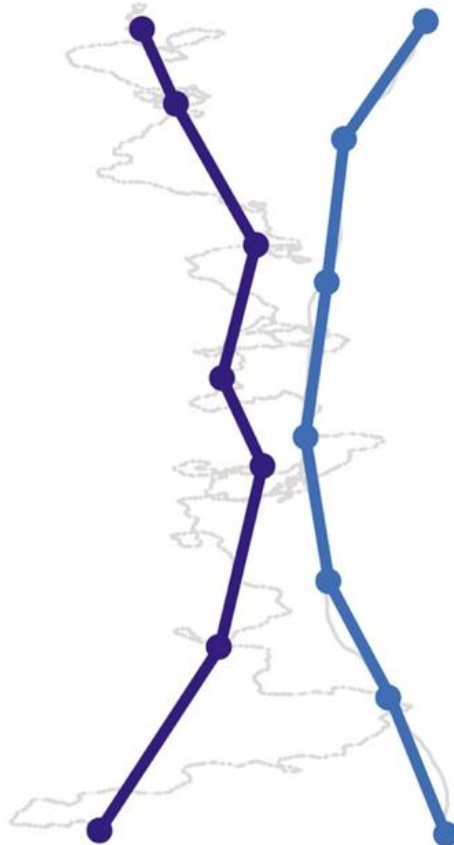
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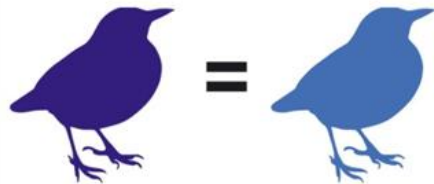
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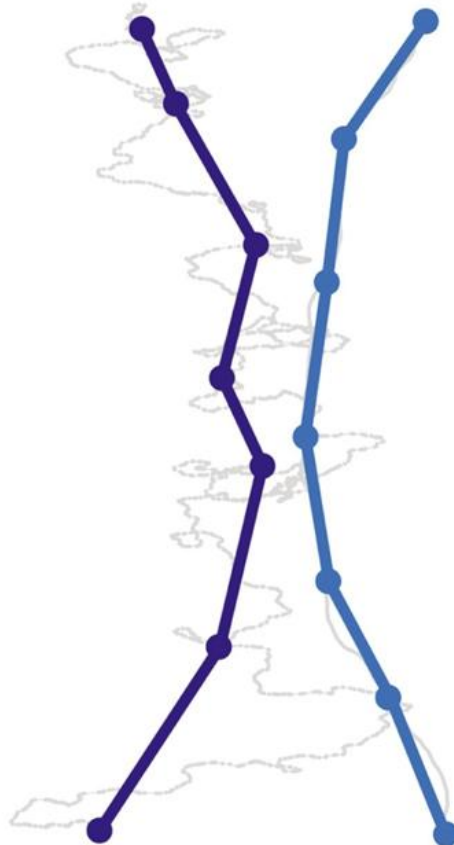
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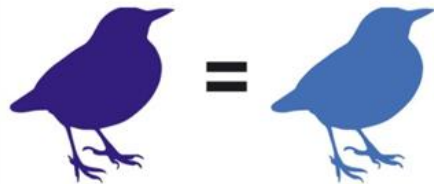
Solutions:

Encounter estimation & prediction

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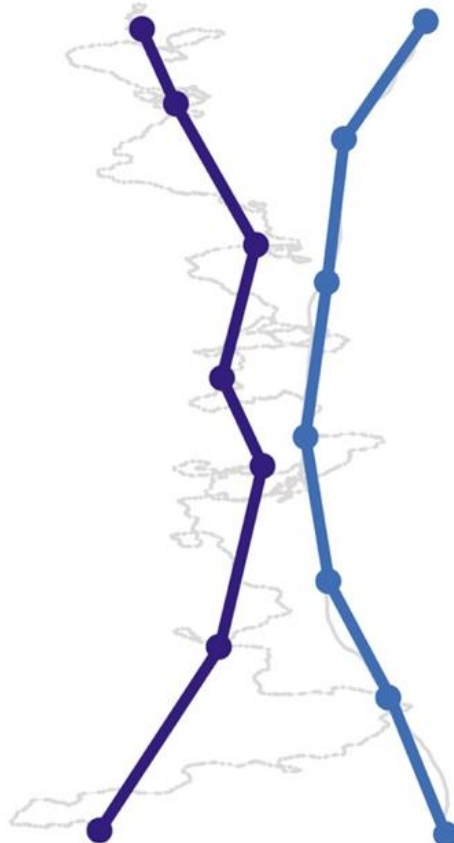
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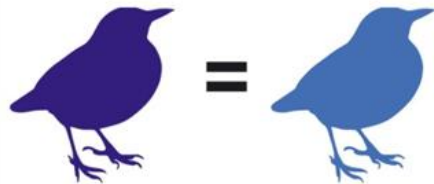
- Increased sampling rate

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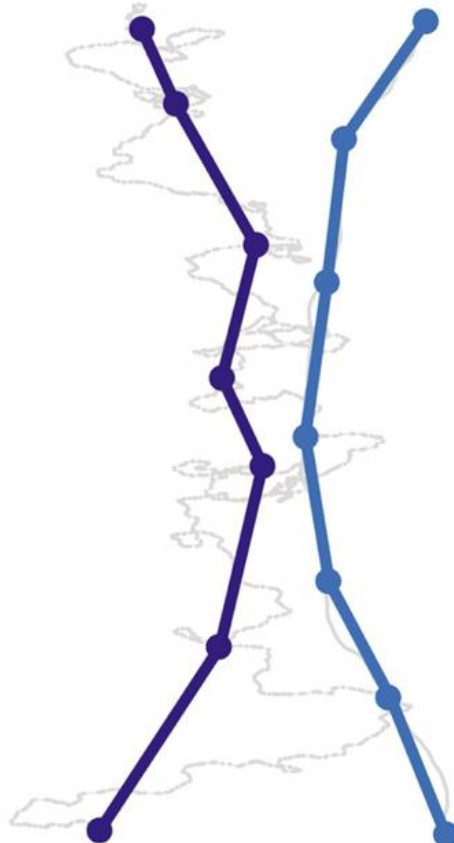
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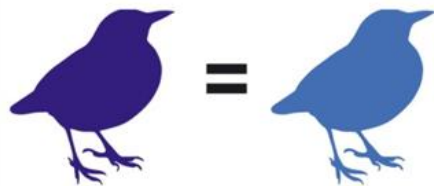
- Increased sampling rate
- **Continuous-time** stochastic process models

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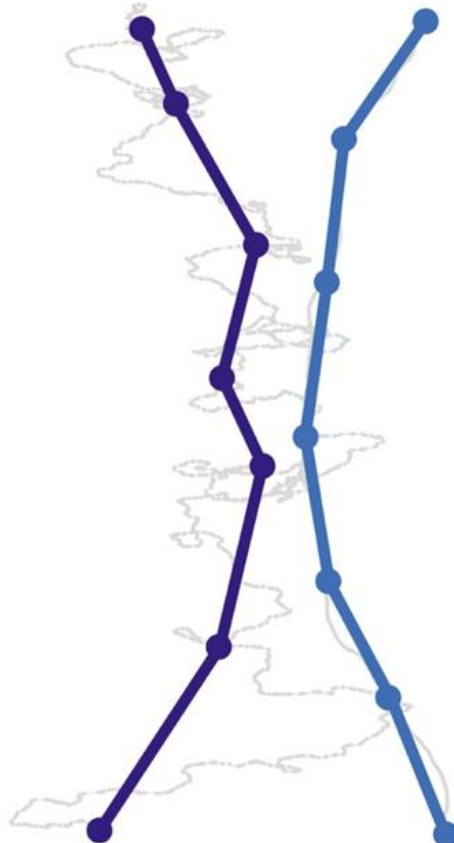
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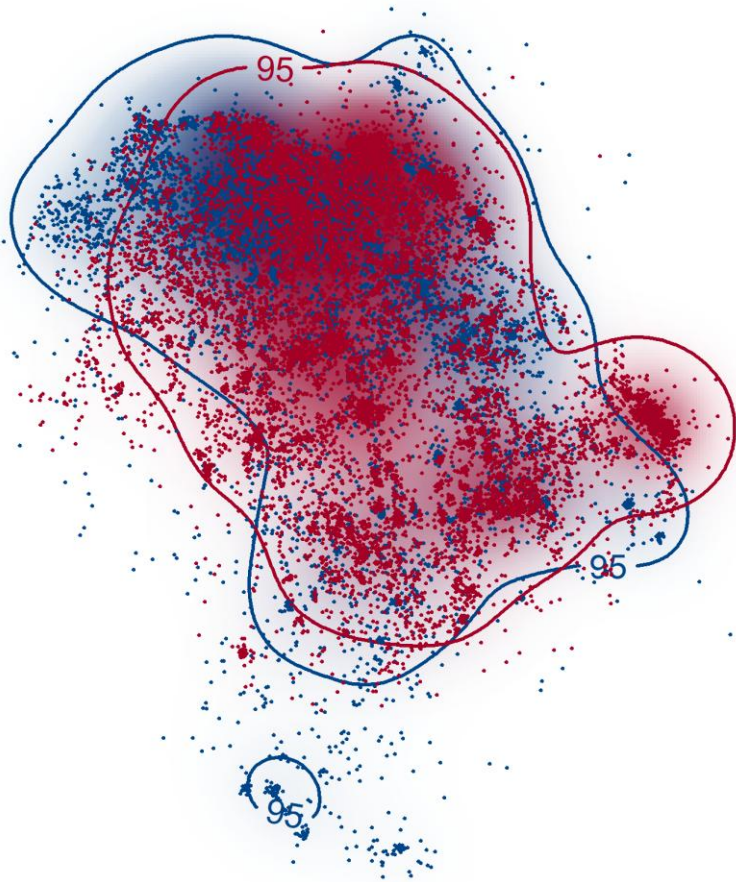
Solutions:

- Increased sampling rate
- **Continuous-time** stochastic process models & model-based inference

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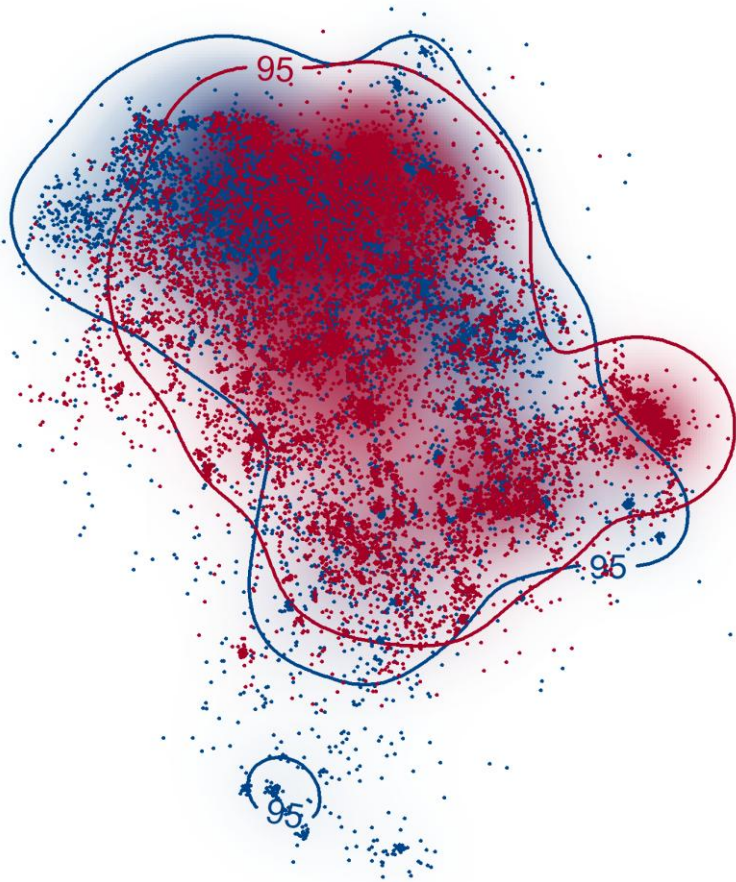
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Spatio-temporal interactions



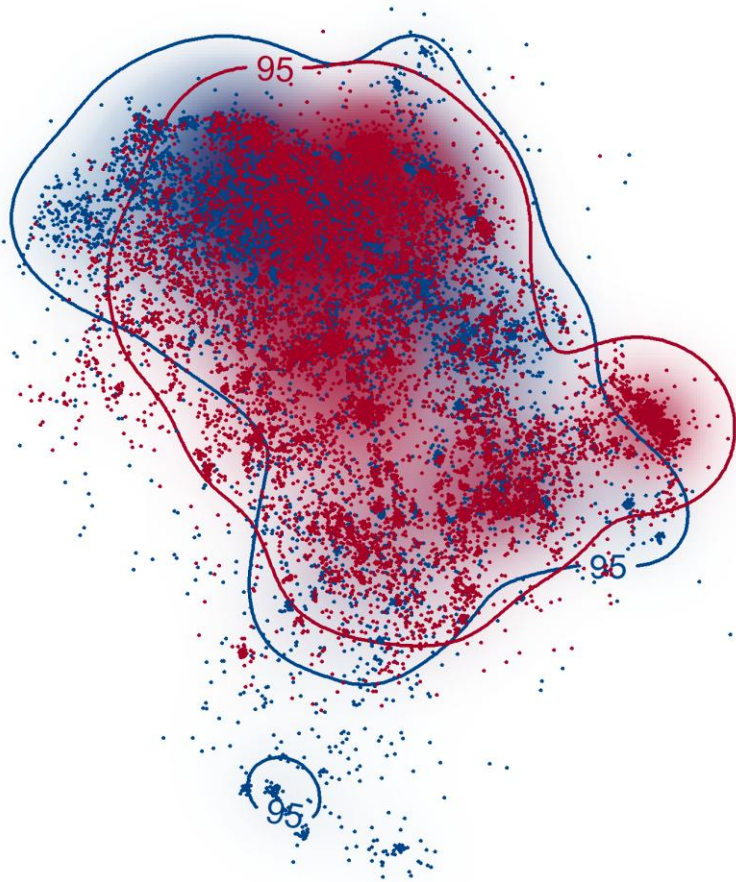
- Home-range overlap describes the *extent to which animals share the same space*

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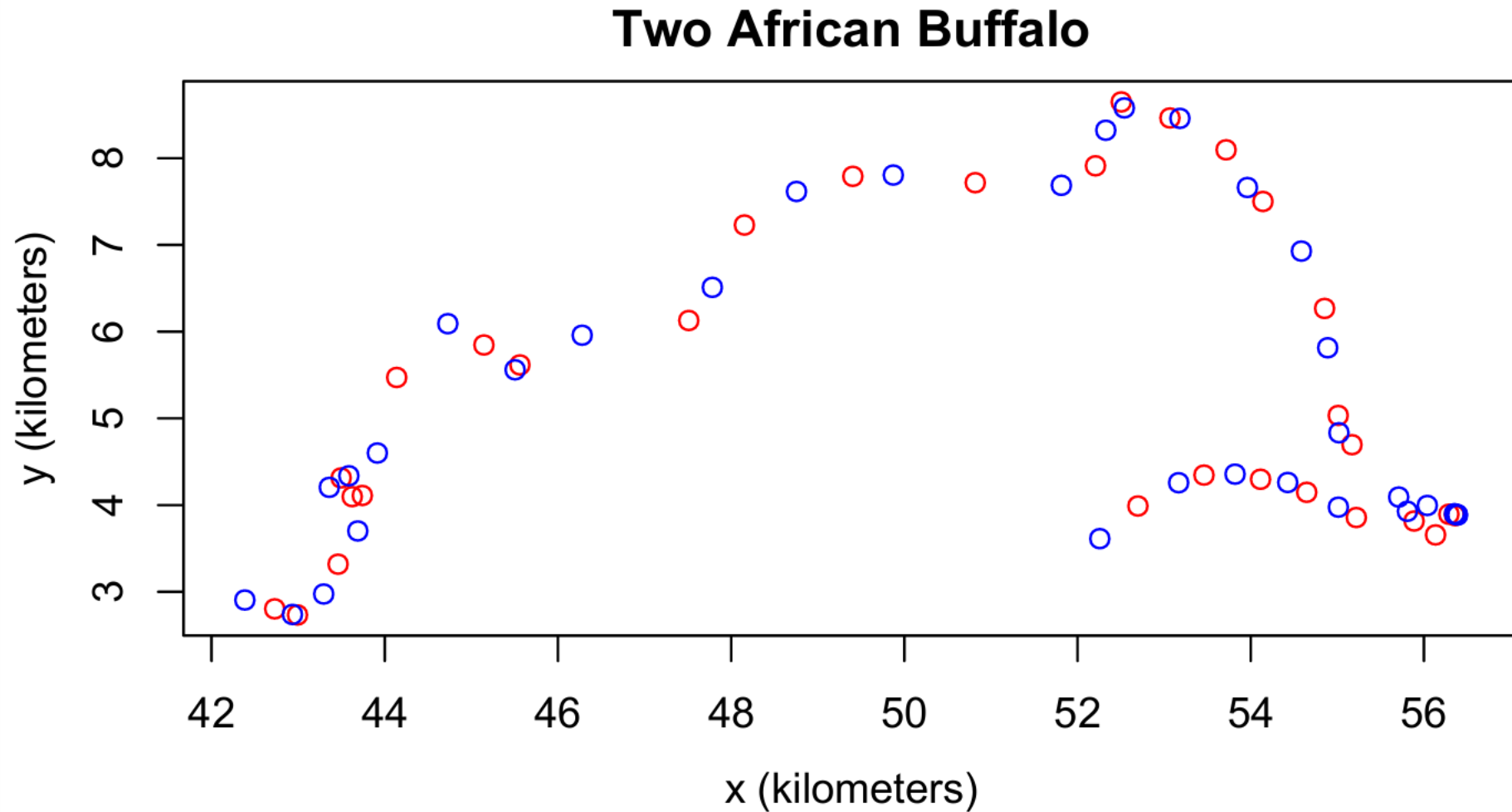
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Spatio-temporal interactions



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 - But says nothing about whether animals share the **same space at the same time**
- **Pairwise separation distances** are a good place to start for understanding if pairs of individuals are **attracted** to, or **avoid**, one another

A simple example:



Pairwise distance estimation

Pairwise distance estimation

- 1) Fit a suite of candidate models:
 - Continuous-time stochastic process models

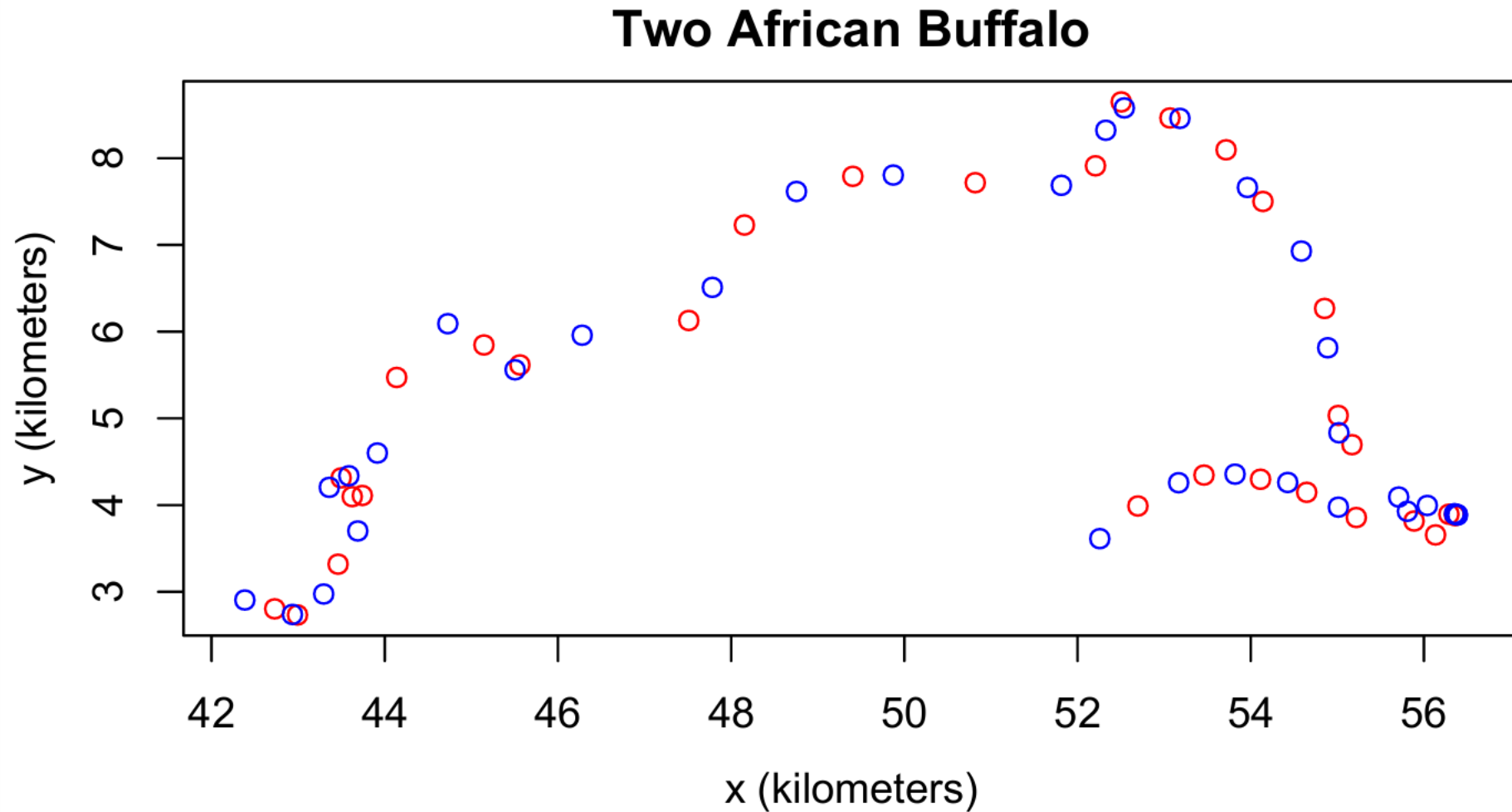
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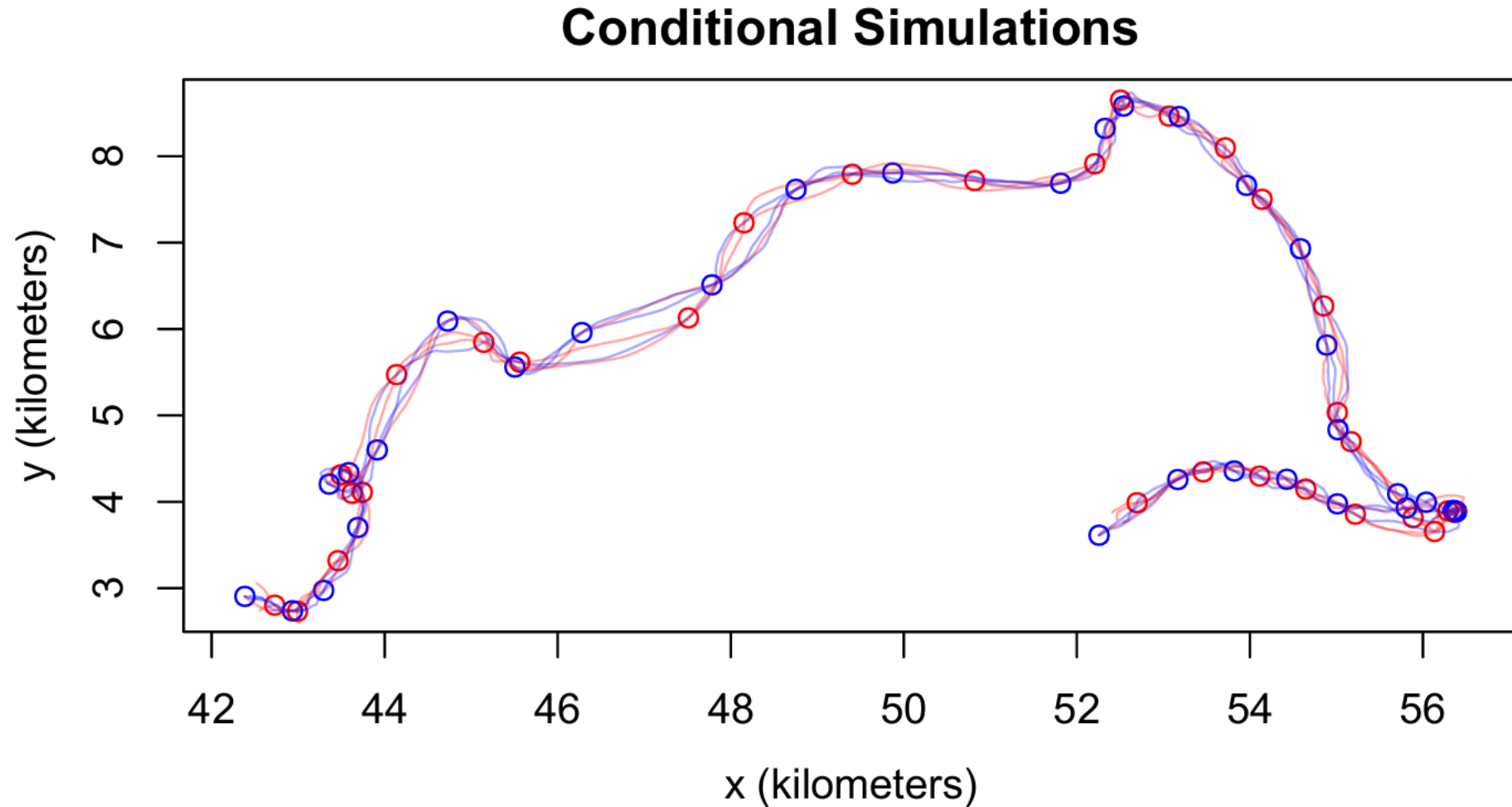
Pairwise distance estimation

- 1) Fit a suite of candidate models:
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- 2) Select the best model
- 3) Model-based inference (which includes conditional simulation)

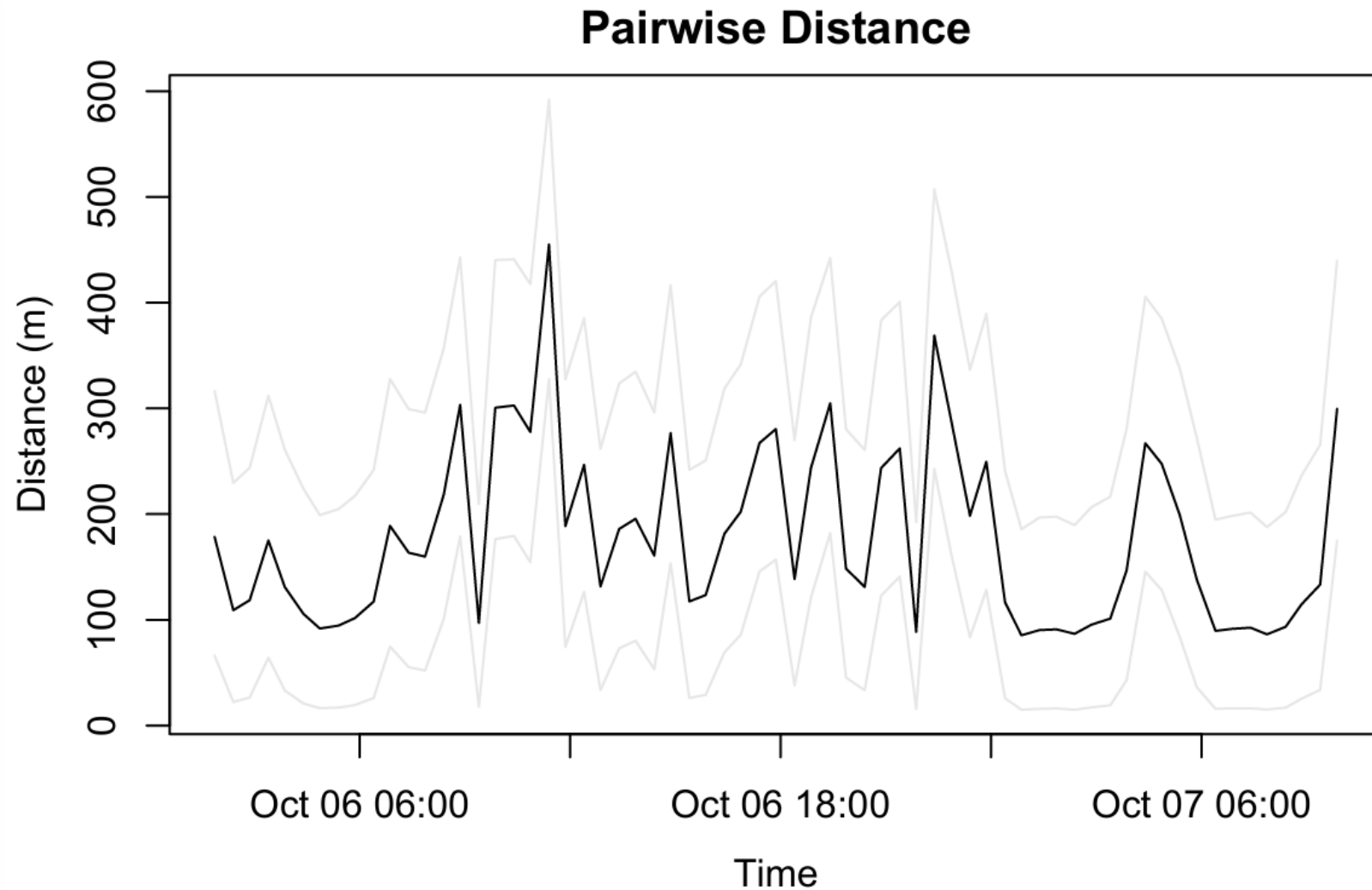
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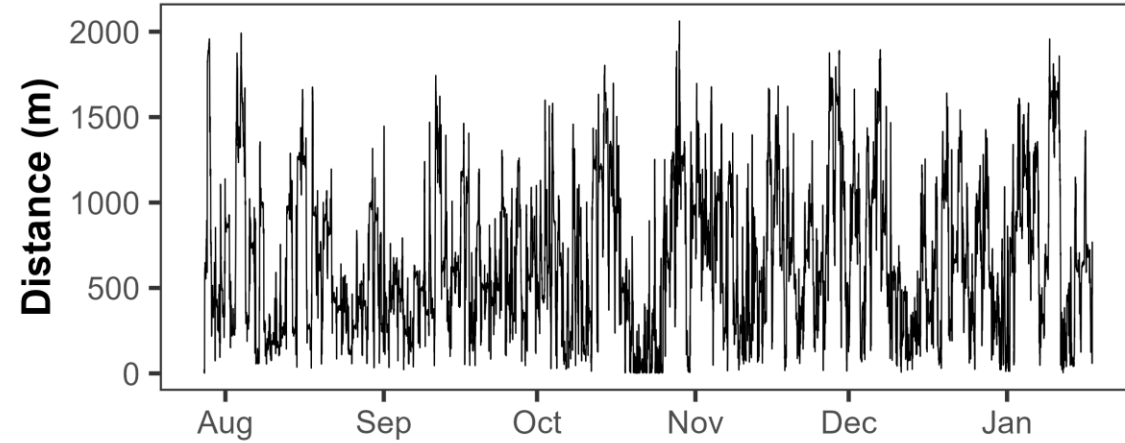
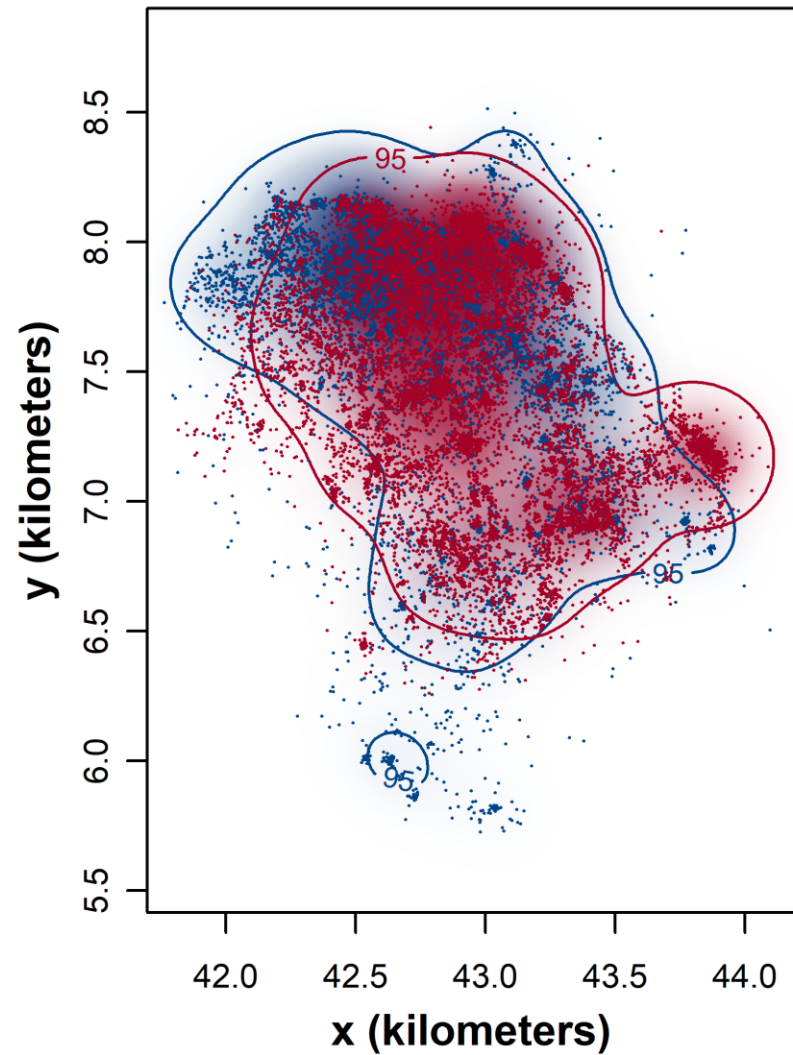


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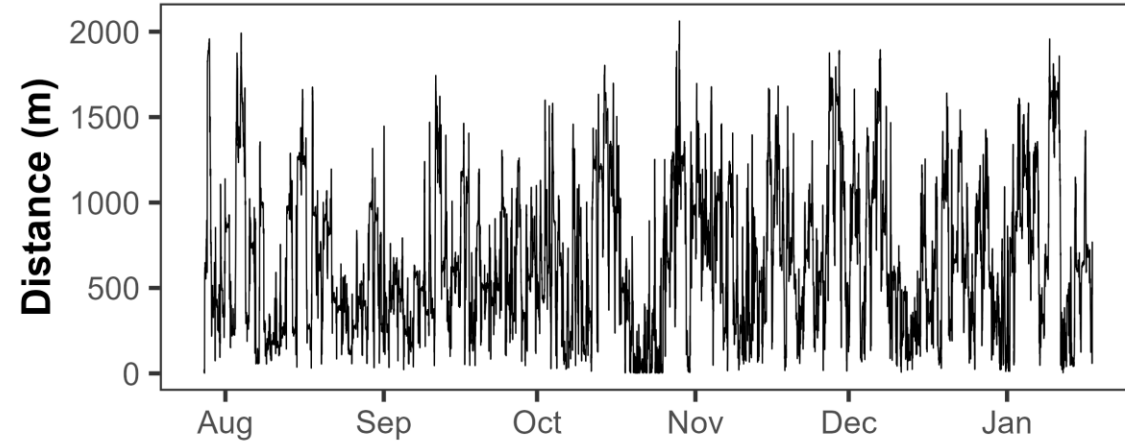
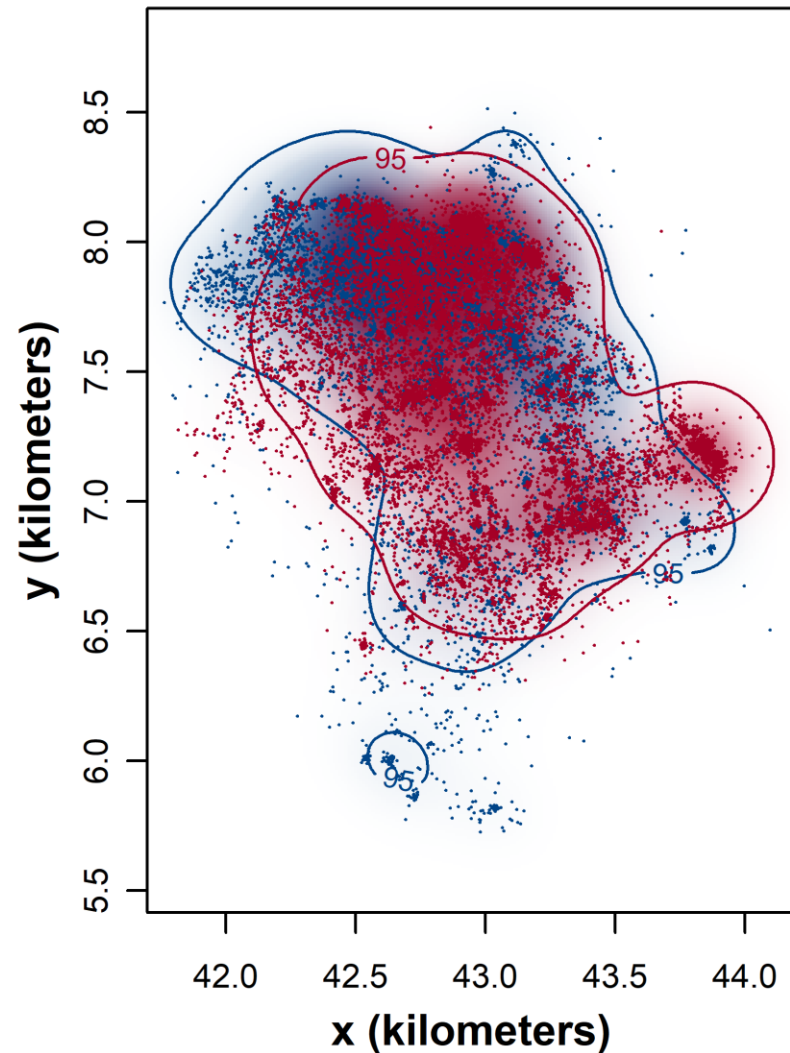
Proximity ratio

Chhen *et al.* (2024)



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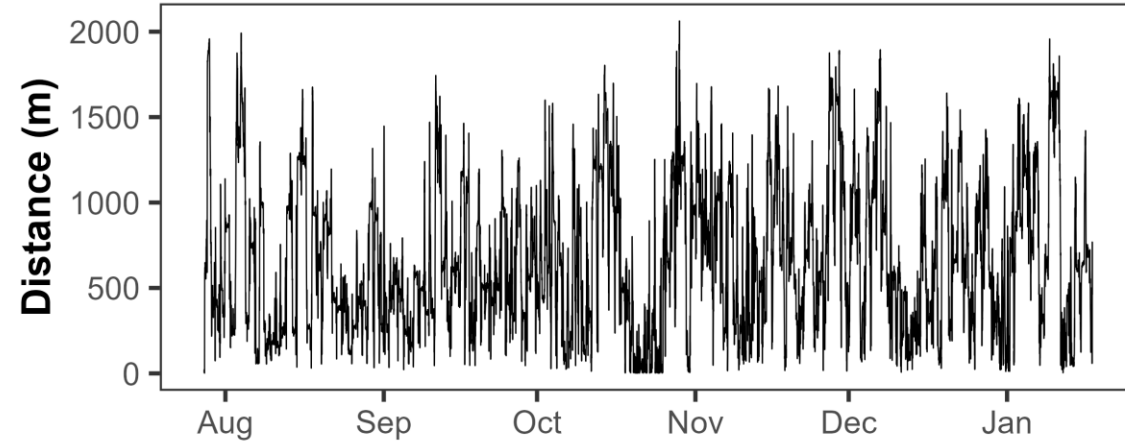
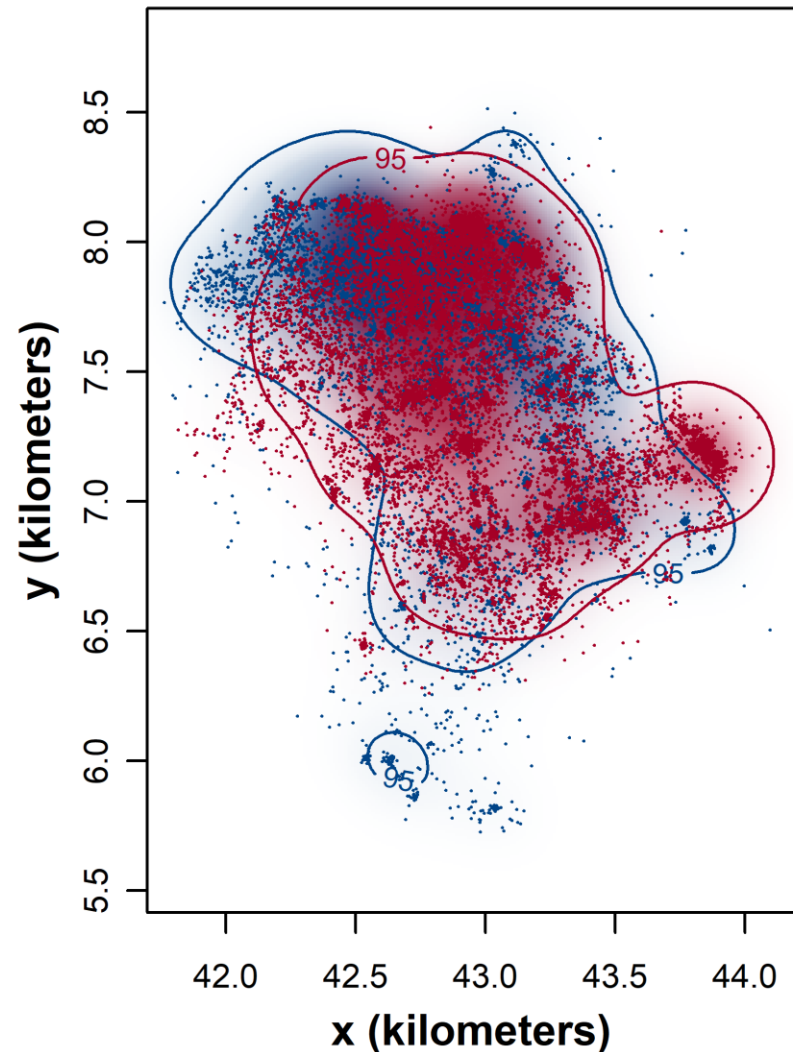
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- Separation distances can provide useful descriptions of pairs of animals

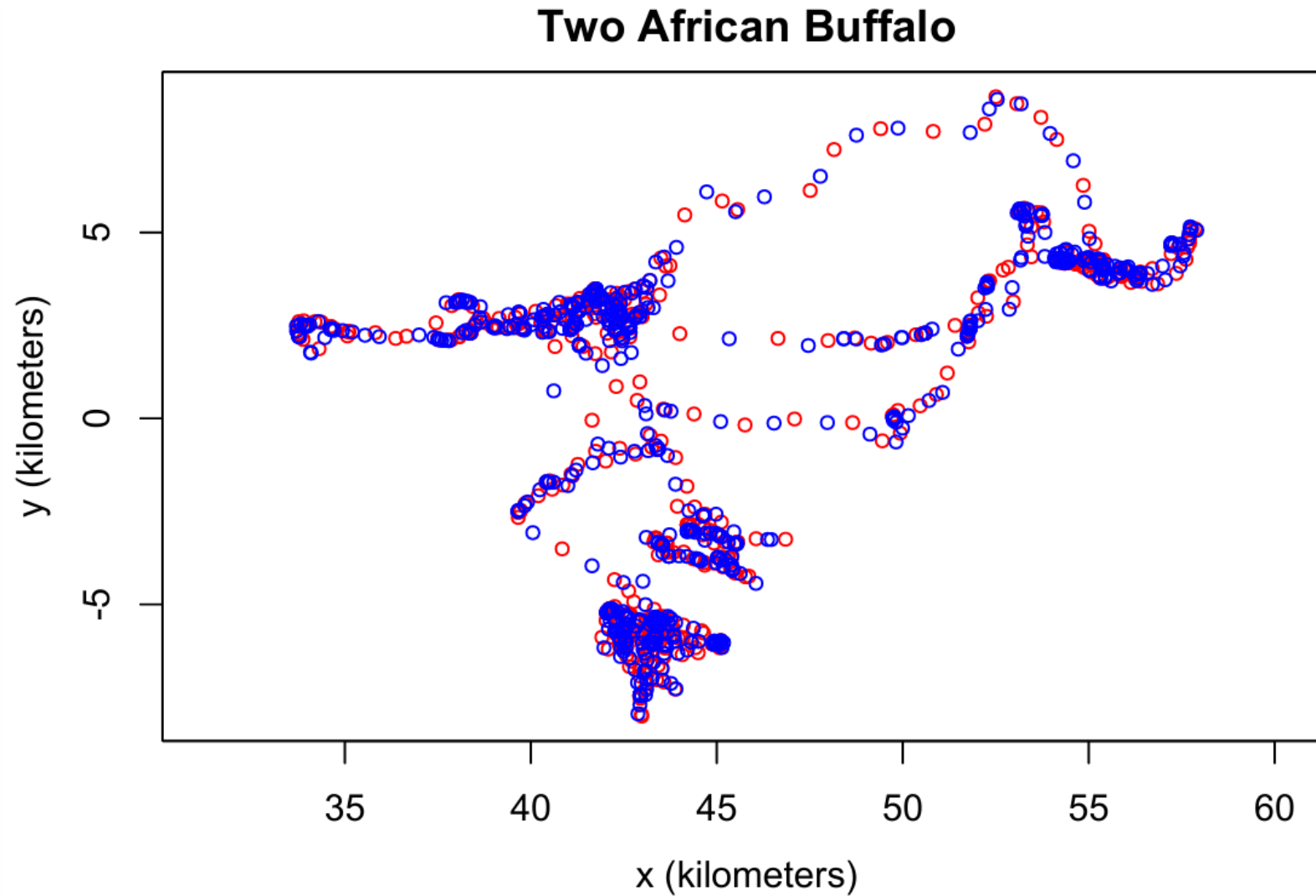
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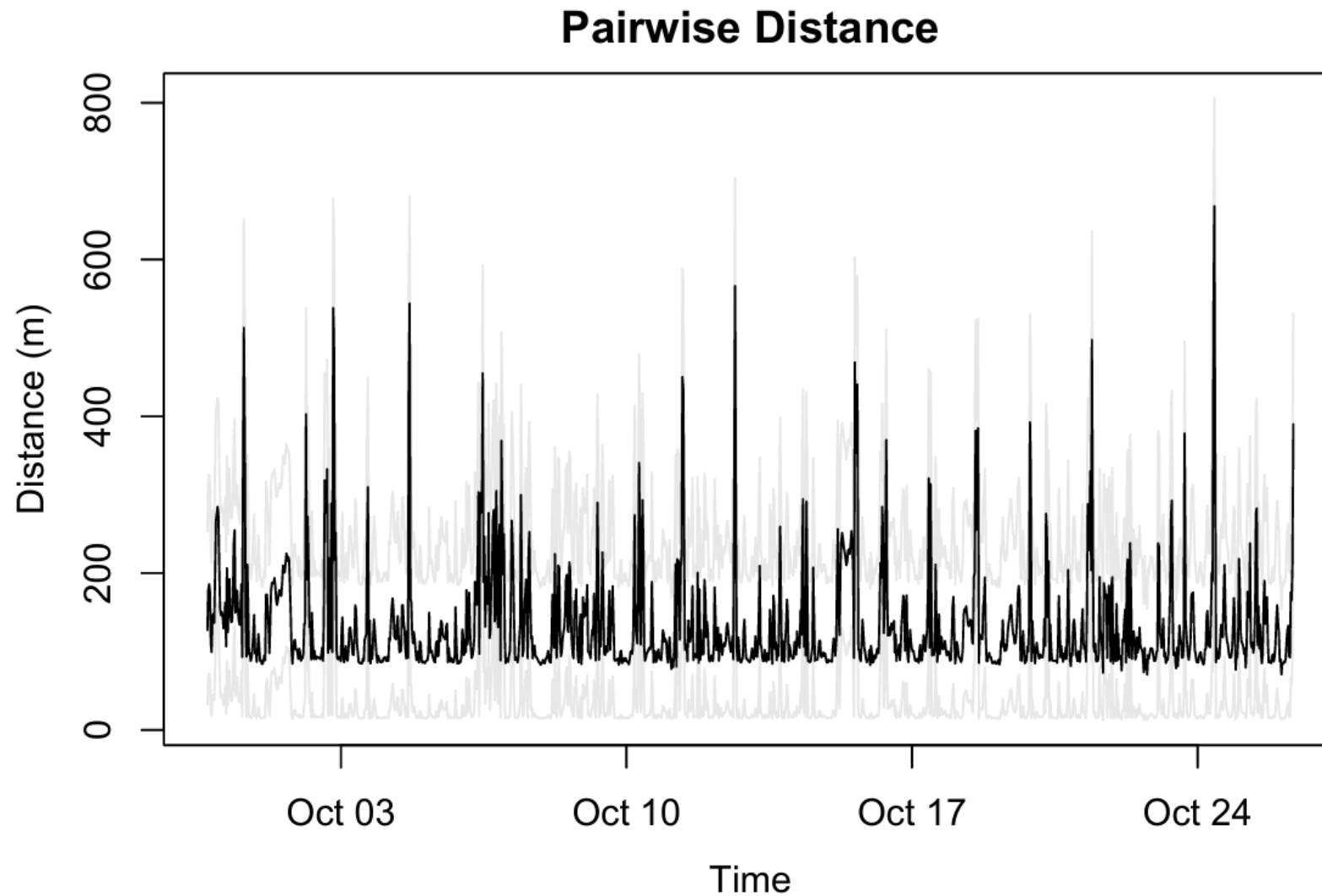


- Separation distances can provide useful descriptions of pairs of animals
 - But are these meaningful?
 - What are they saying about relationships?

Proximity ratio

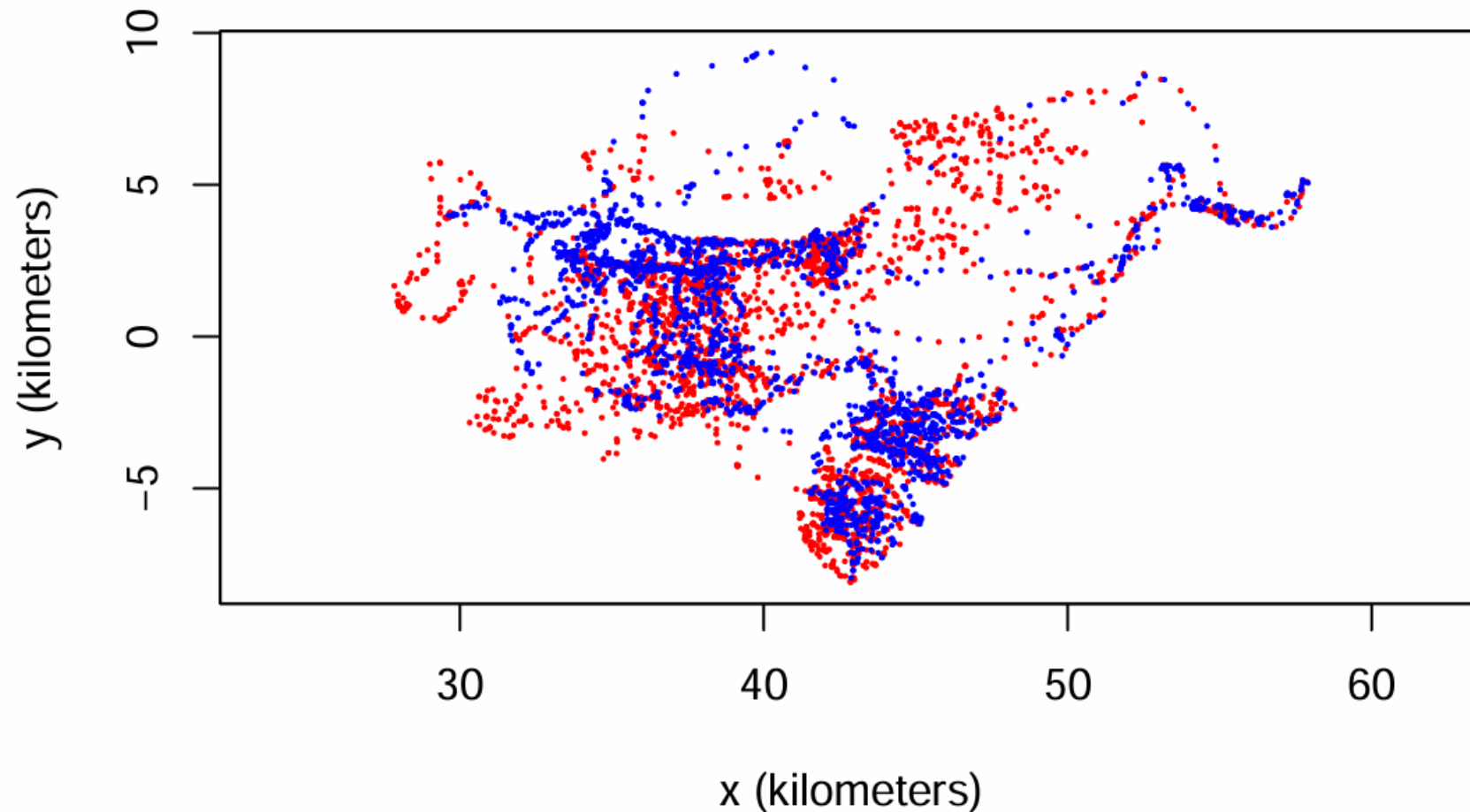


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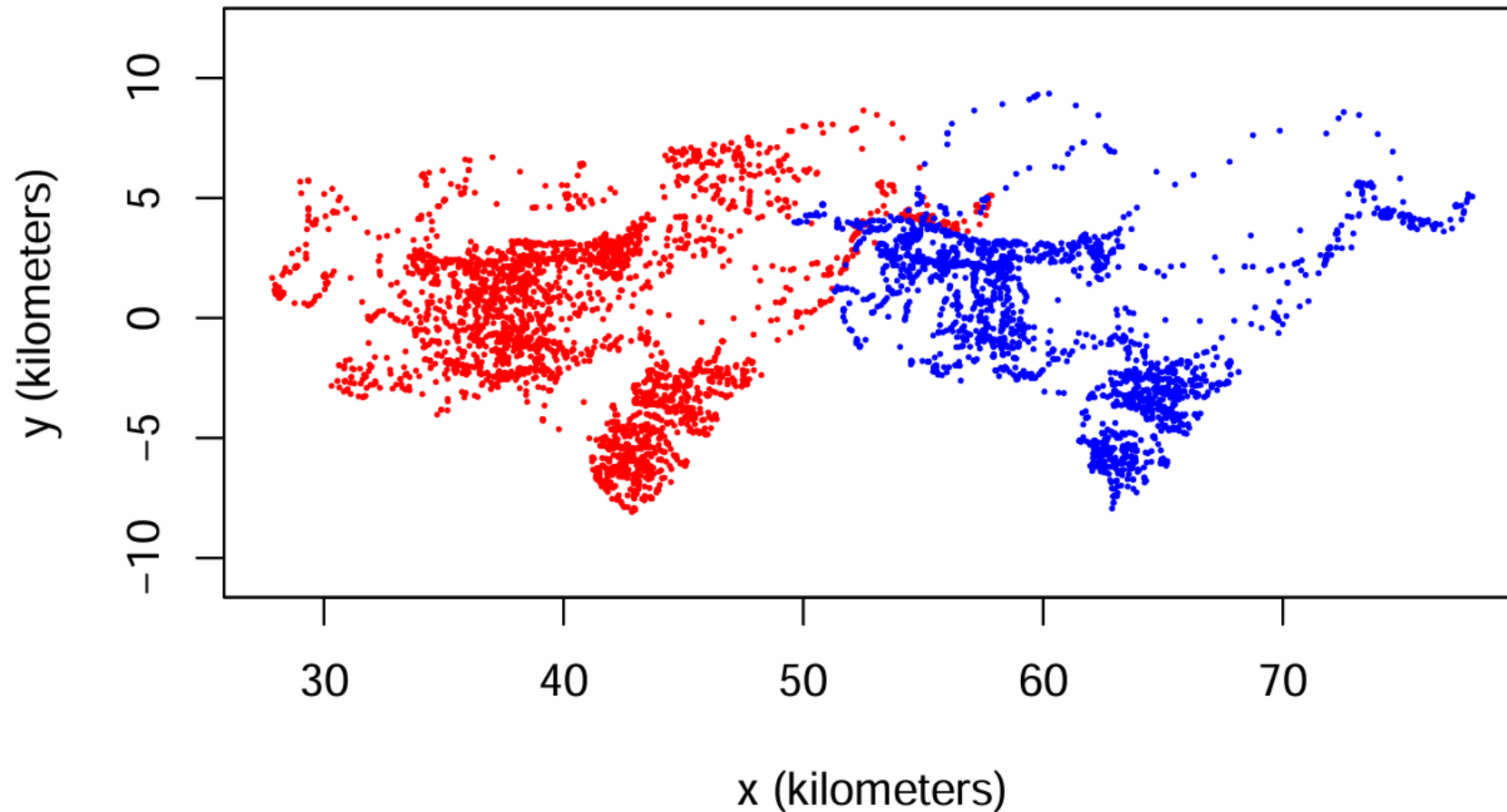
Proximity ratio: Motivation

Positive correlation = Encounters



Proximity ratio: Motivation

Negative correlation = Encounters



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- Construct a simple statistic based on the difference vector $\mathbf{r}_1(t) - \mathbf{r}_2(t)$ that accounts for non-independence and does not involve thresholds.

$$(1) \quad F = \frac{\mathbb{E}[|\mathbf{r}_1(t) - \mathbf{r}_2(t)|^2]}{\mathbb{E}[\mathbf{r}_1(t)^2] - 2 \mathbb{E}[\mathbf{r}_1(t)] \mathbb{E}[\mathbf{r}_2(t)] + \mathbb{E}[\mathbf{r}_2(t)^2]}$$

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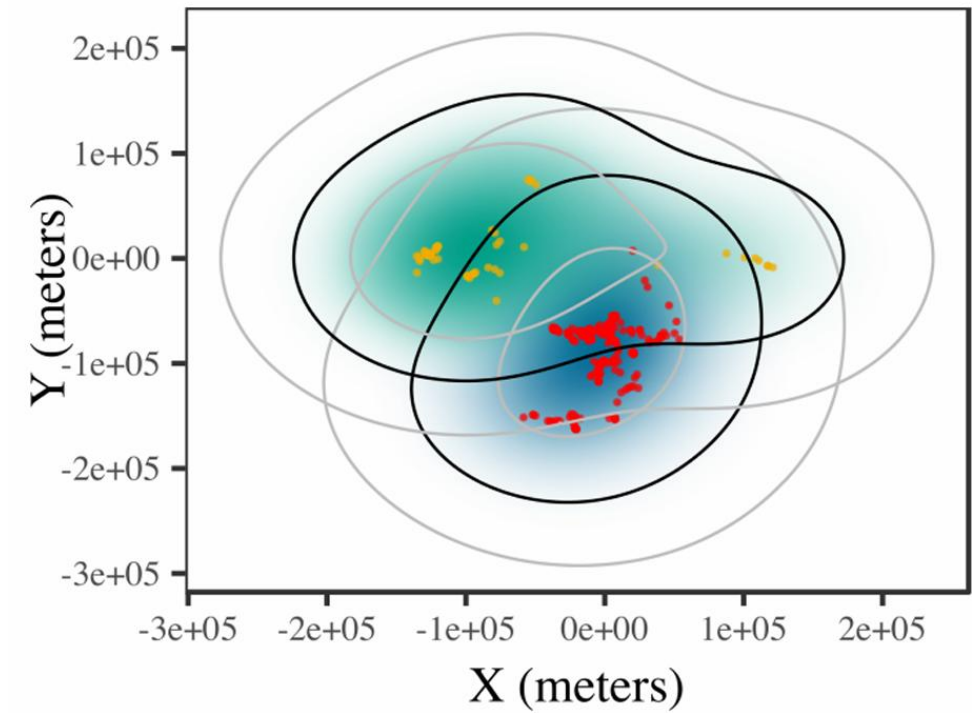
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- Buffalo example: $\hat{F} = 0.012\%$ (0.0085–0.018%)

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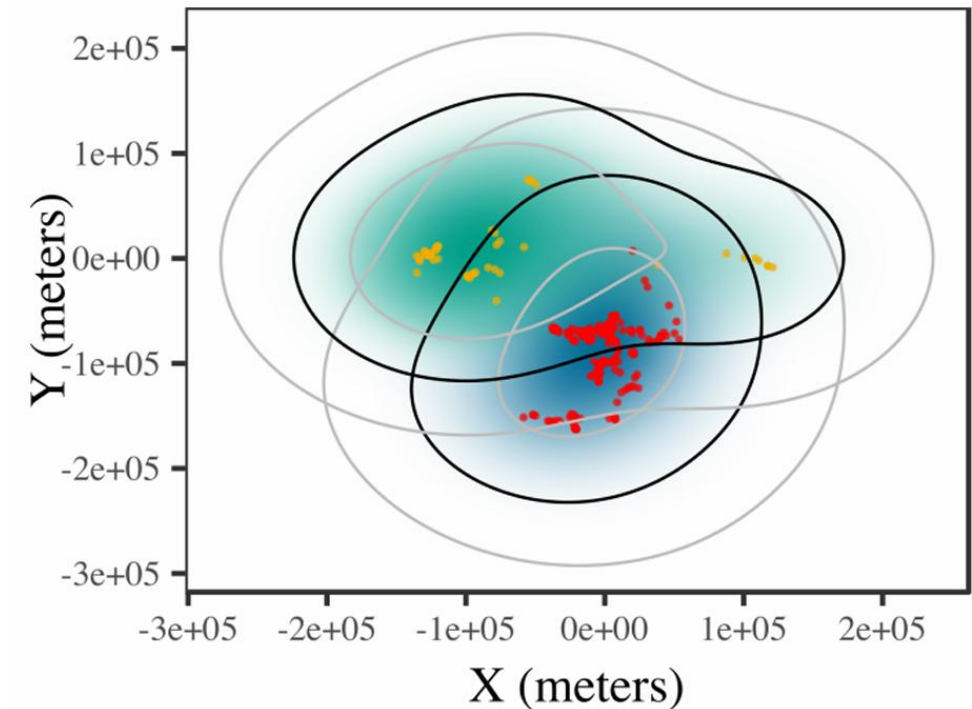
Home-range overlap

- Home ranges quantify typical space needs



Home-range overlap

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- **HR overlap** of home ranges captures **potential for interactions**
 - Many different measures used in practice
 - Reviewed by Fieberg & Kochanny, 2005



Statistical inference for overlap

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- 1) Comparable home range estimates

Statistical inference for overlap

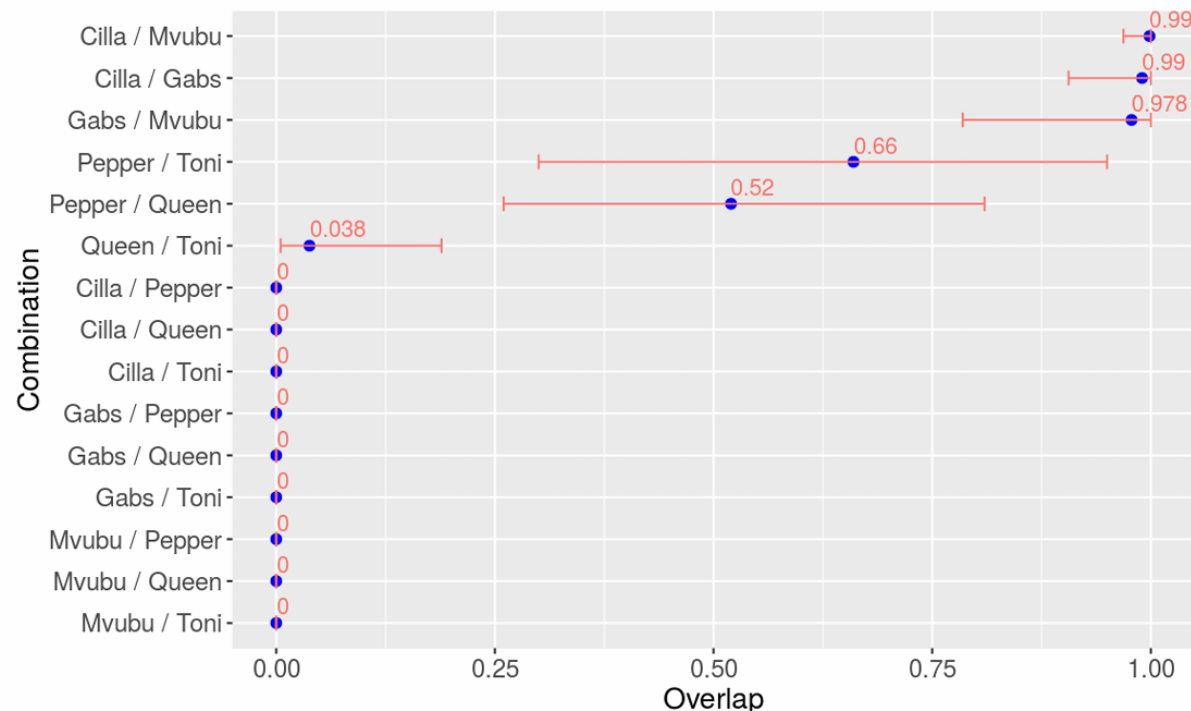
- 1) Comparable home range estimates
- 2) An interpretable, relatively unbiased overlap index

Statistical inference for overlap

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- 3) A way to propagate uncertainty in HR estimates into overlap estimation

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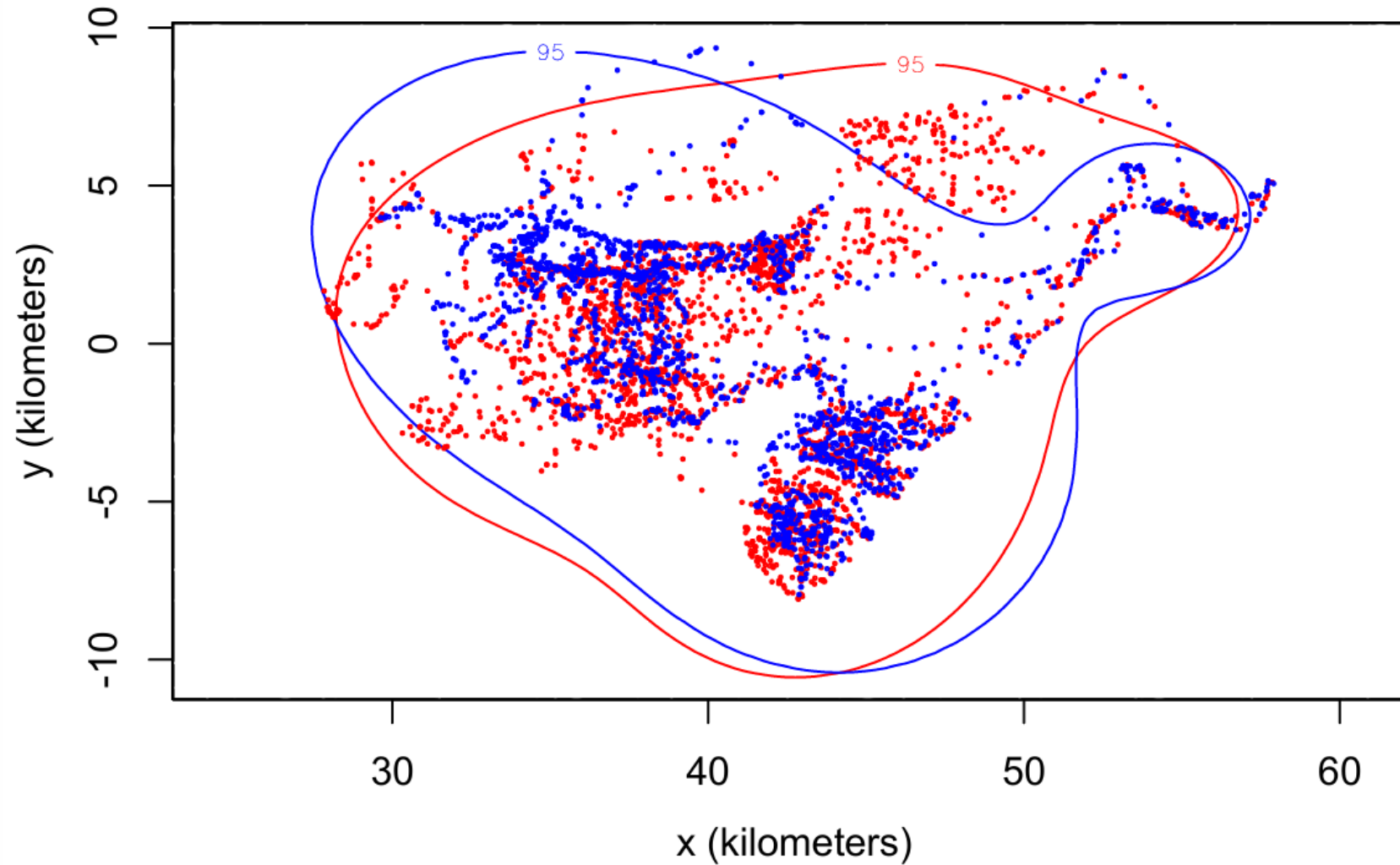
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(Conventional)

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Home-range overlap: Conventional



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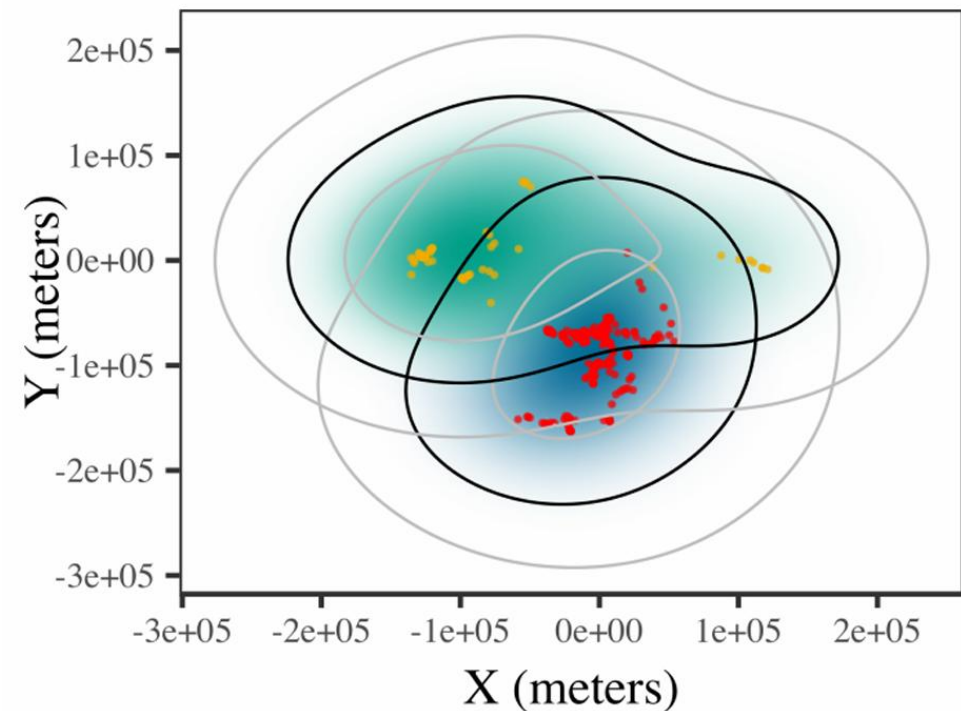
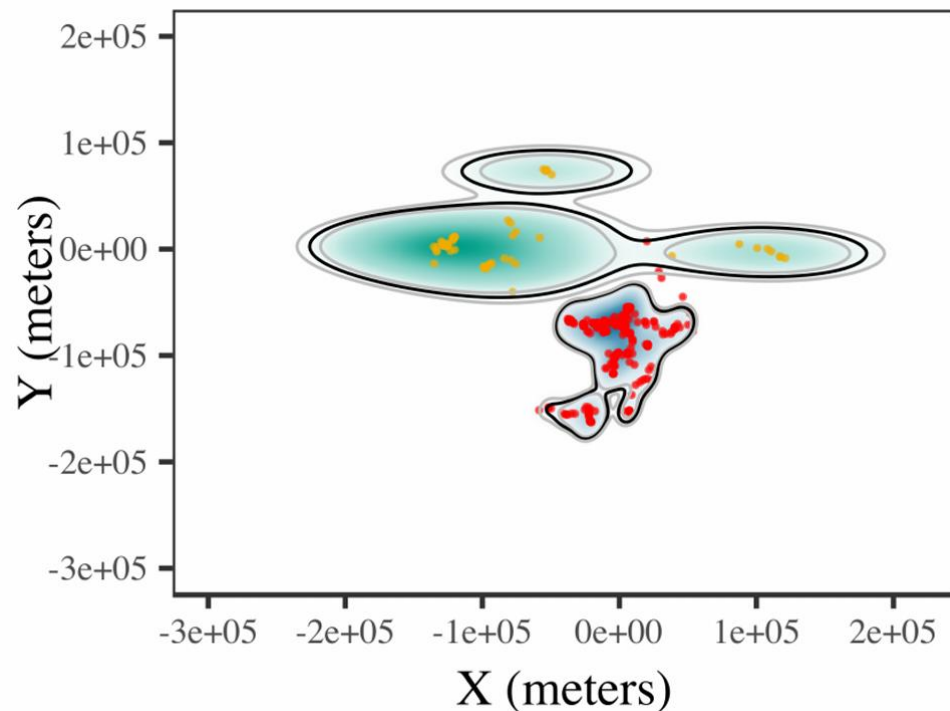
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$$\text{overlap}_{12}(P) = \frac{\iint_{\mathbb{R}^2} d^2\mathbf{r} p_1(\mathbf{r}) p_2(\mathbf{r})}{\sqrt{\iint_{\mathbb{R}^2} d^2\mathbf{r}' p_1(\mathbf{r}')^2 \iint_{\mathbb{R}^2} d^2\mathbf{r}'' p_2(\mathbf{r}'')^2}}$$

Statistical inference for overlap

Bias in HR estimation **propagates into bias in overlap**



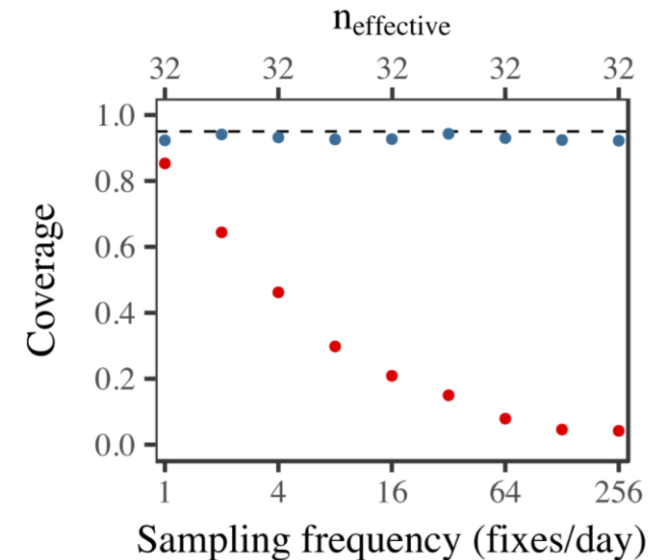
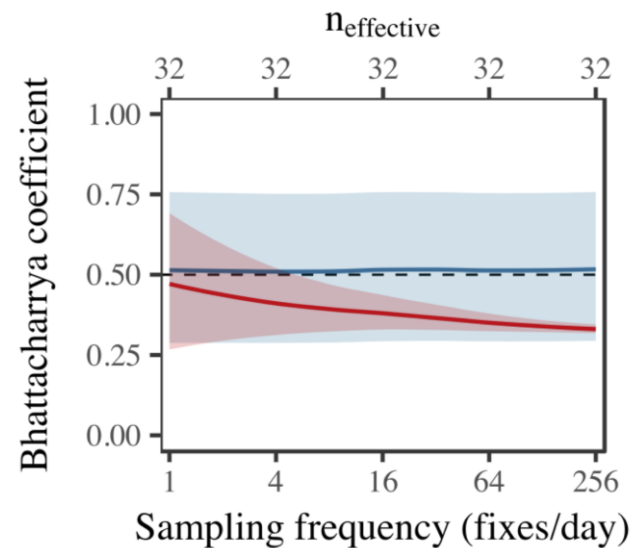
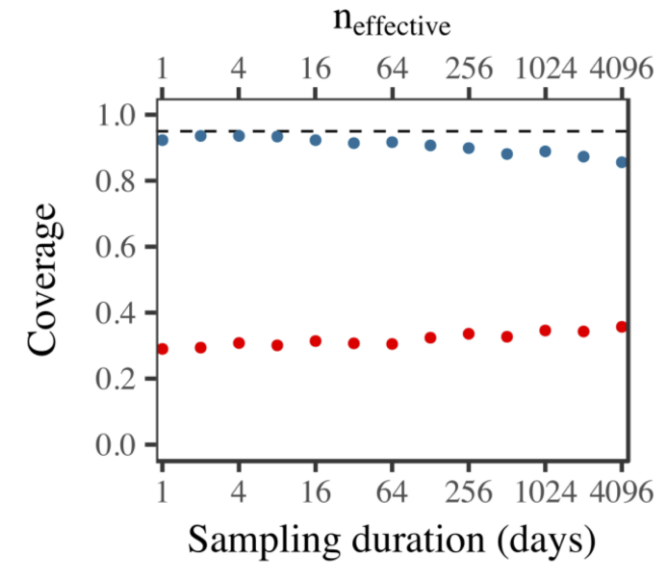
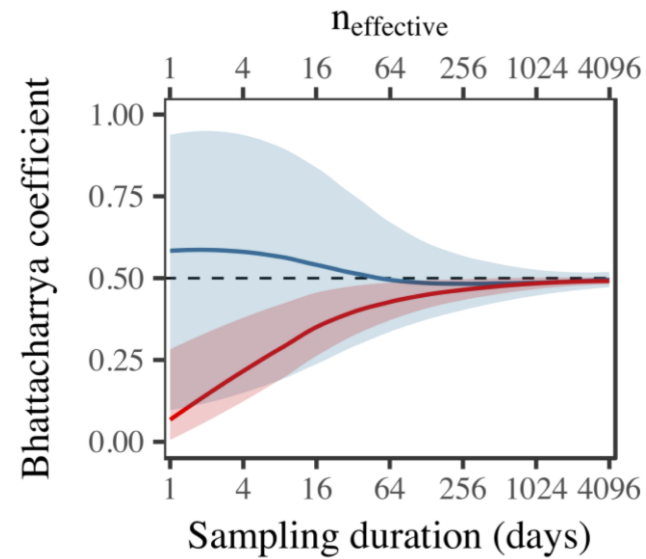
Statistical inference for overlap

$$D_B(p_1, p_2) = \frac{1}{8}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^T \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2) + \frac{1}{2} \ln \left(\frac{\det \boldsymbol{\Sigma}}{\sqrt{\det \boldsymbol{\Sigma}_1 \det \boldsymbol{\Sigma}_2}} \right)$$

$$BC = e^{-D_B(p_1, p_2)}$$

- Measure of overlap of two probability distributions
 - Already familiar to, and in use by ecologists
- Mathematically tractable:
 - 1st order bias correction and error propagation possible

Behavior of the estimator



Overview

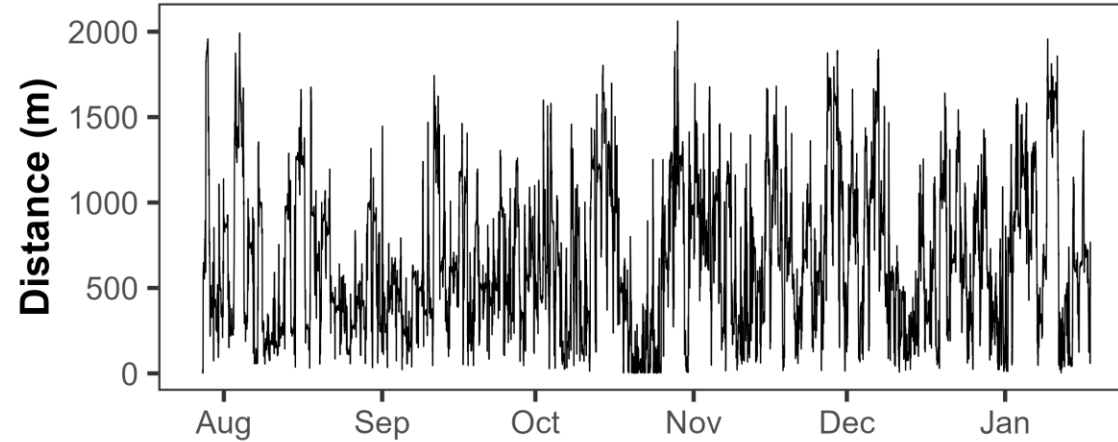
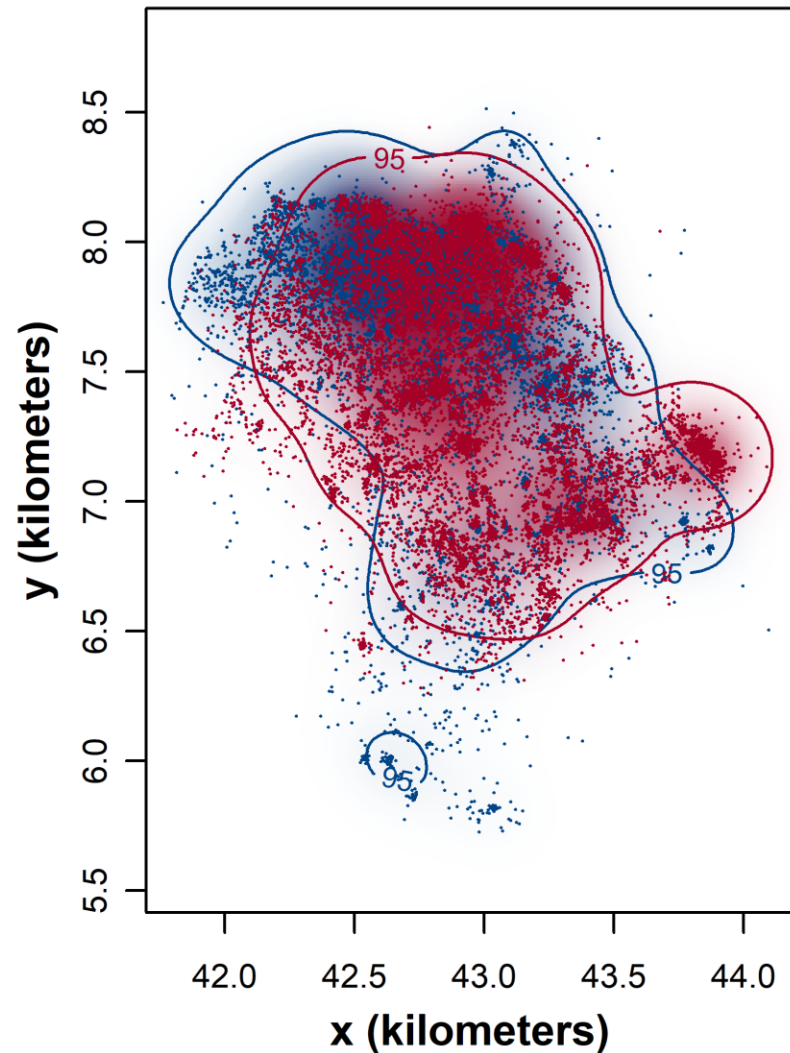
- 1) Introduction
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- 5) Encounter probability & Encounter rates**
- 6) Conditional distribution of encounters
- 7) Future directions

Encounter rates

- We're often interested in linking individual behavior with higher-level ecological processes
- For many processes, this is governed by how an individual's movement behavior translates to *encounter rates*

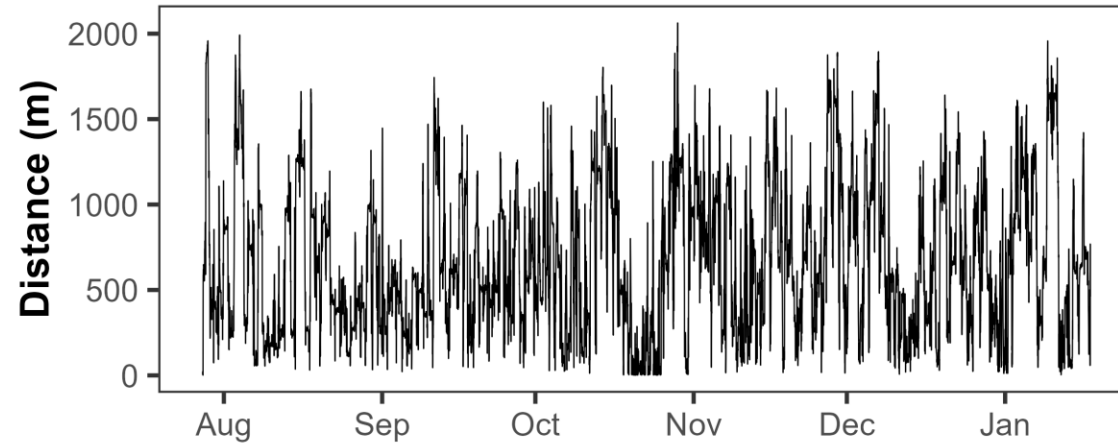
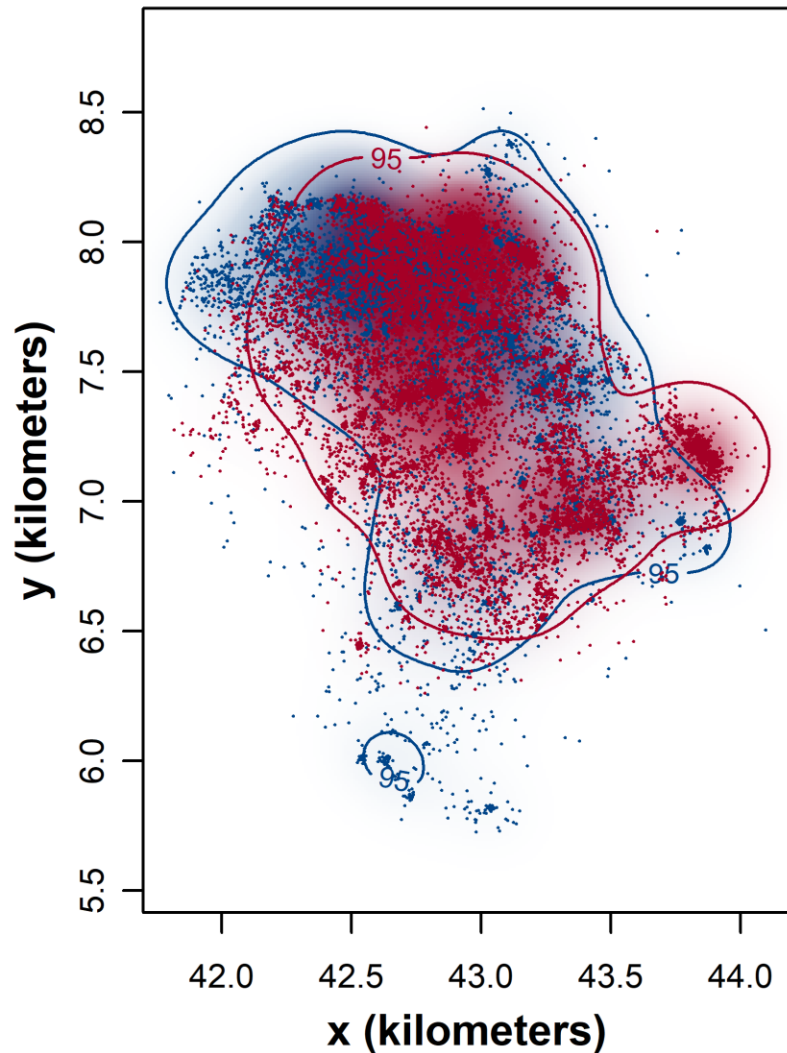


Encounter rates



- The observed separation distances contain information on encounter events (and by extension, encounter rates)

Encounter rates



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 - But are dependent on **individuals' perceptual ranges**

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$$P_{12}(r \leq d) = \pi d^2 \underbrace{\iint_{\mathbb{R}^2} d^2 \mathbf{r} p_1(\mathbf{r}) p_2(\mathbf{r})}_{\mu}$$

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 - Gurarie & Ovaskainen, *Theoretical Ecology* (2013)
- Zero infection-rate ($C = 0$) encounter probability μ
 - Kenkre & Sugara, *Bulletin of Mathematical Biology* (2014)

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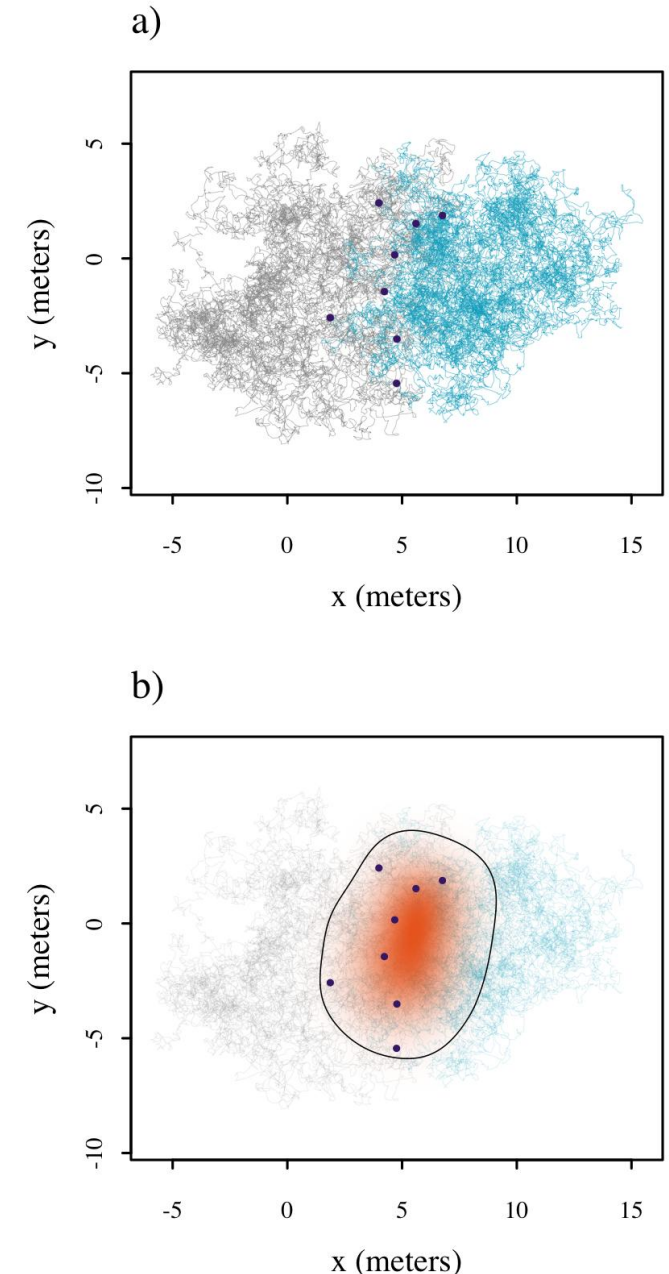
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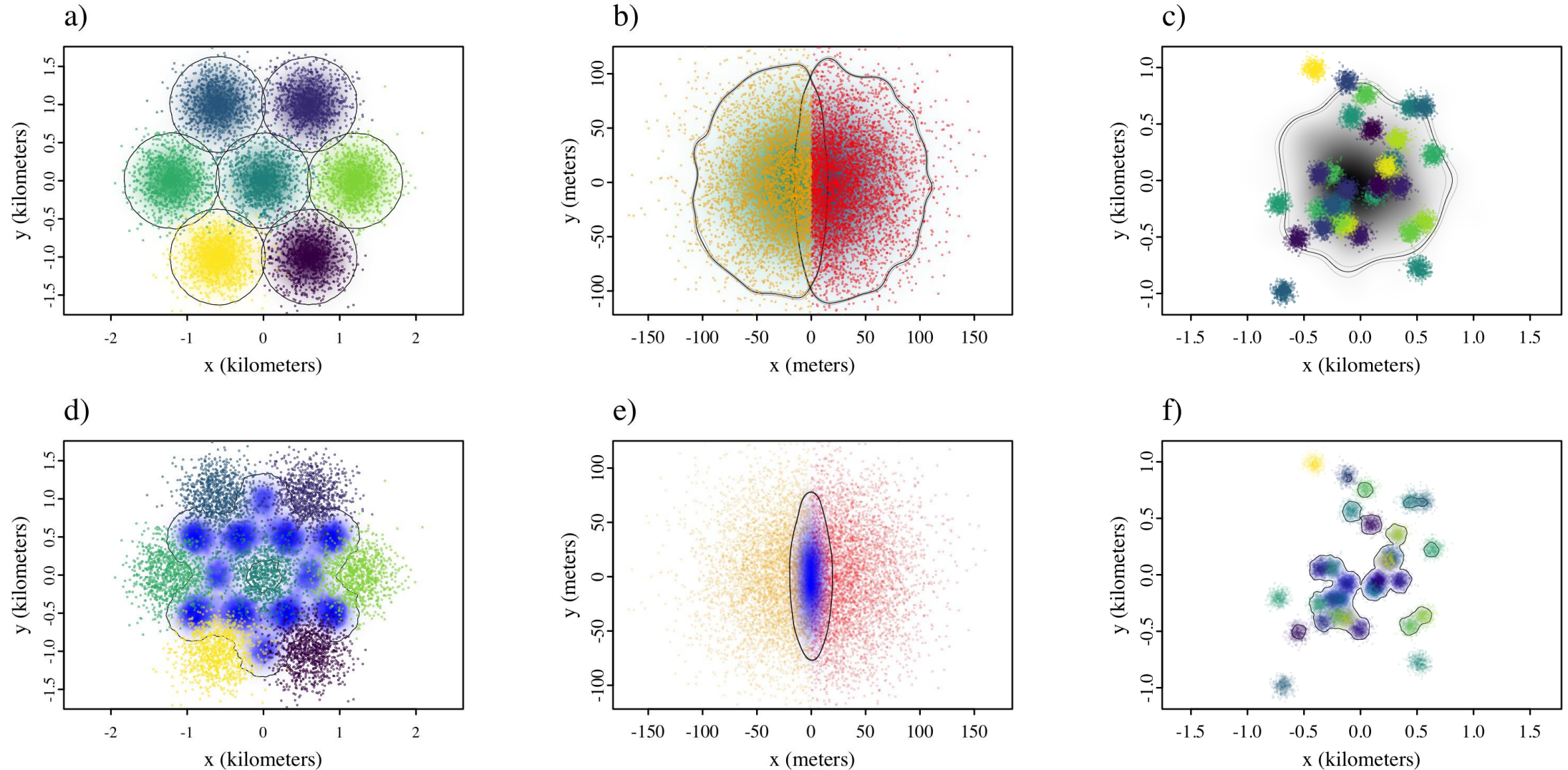
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- The CDE between individuals i and j is given by:

$$\text{CDE}_{ij}(\mathbf{r}) = \frac{p_i(\mathbf{r}) p_j(\mathbf{r})}{\iint d^2\mathbf{r}' p_i(\mathbf{r}') p_j(\mathbf{r}')}$$



Simulation examples

 Noonan *et al.* (2021)



Conditional Distribution of Encounters

- Conditional on an encounter taking place, where did it occur?

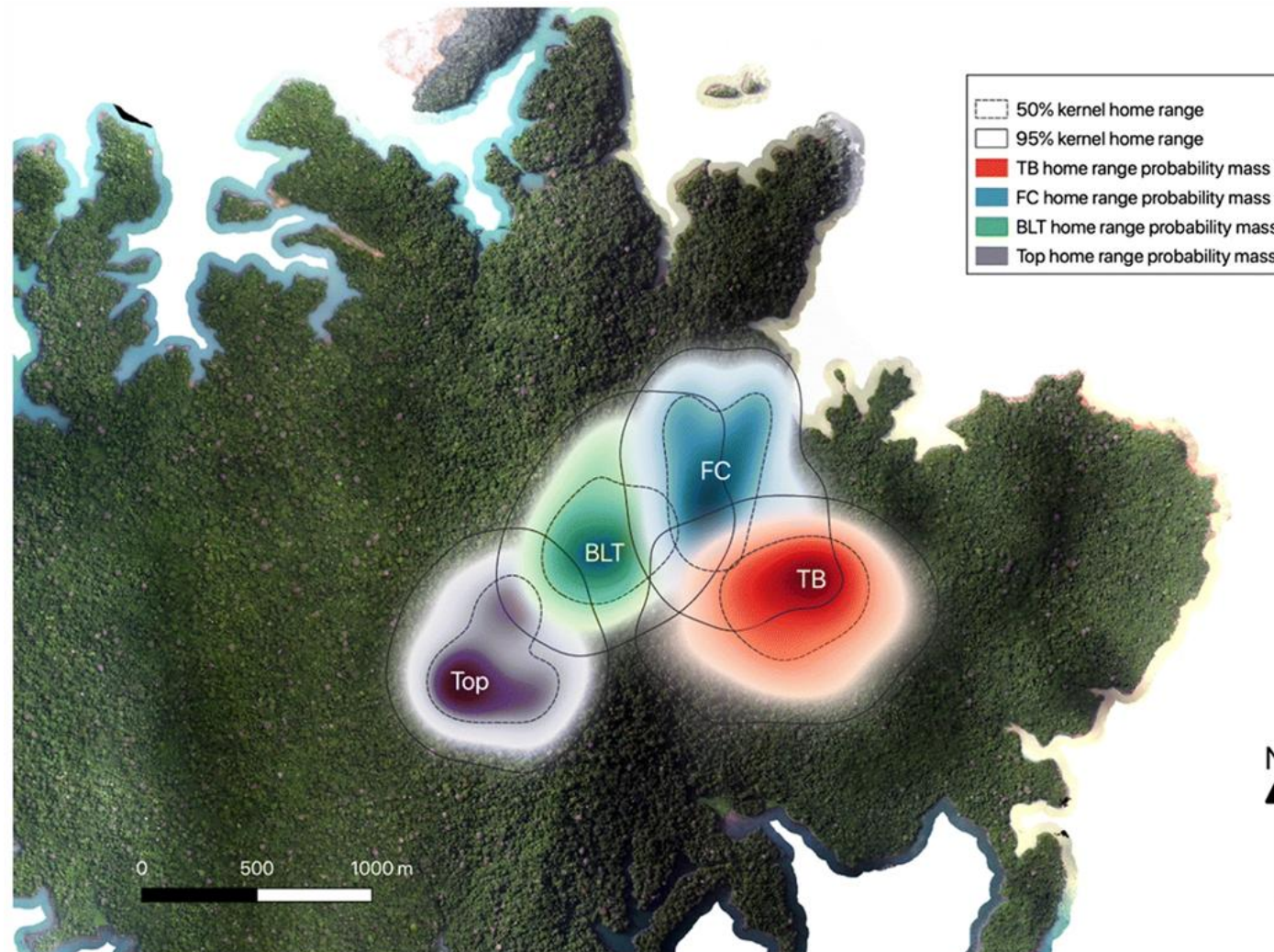
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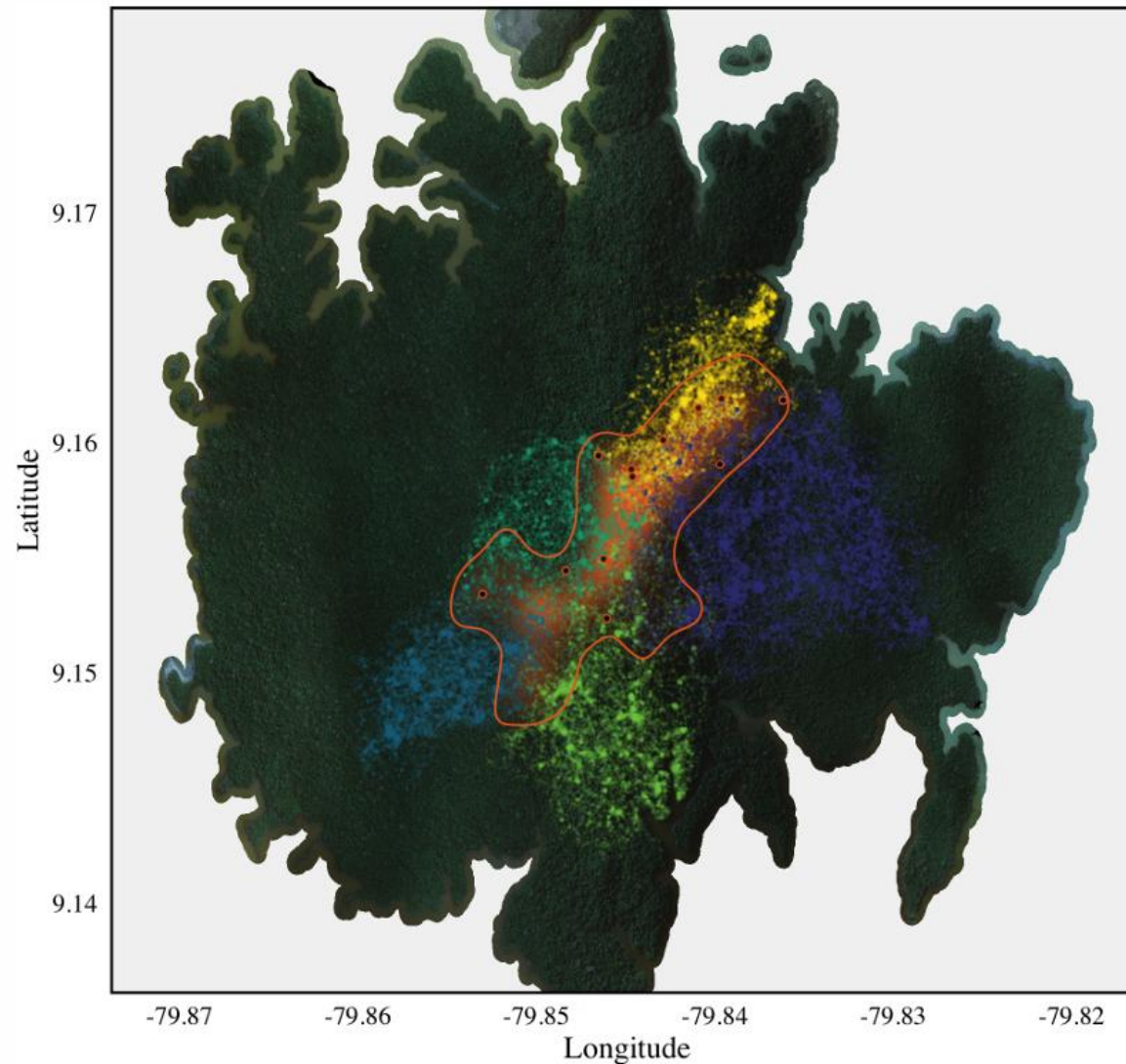
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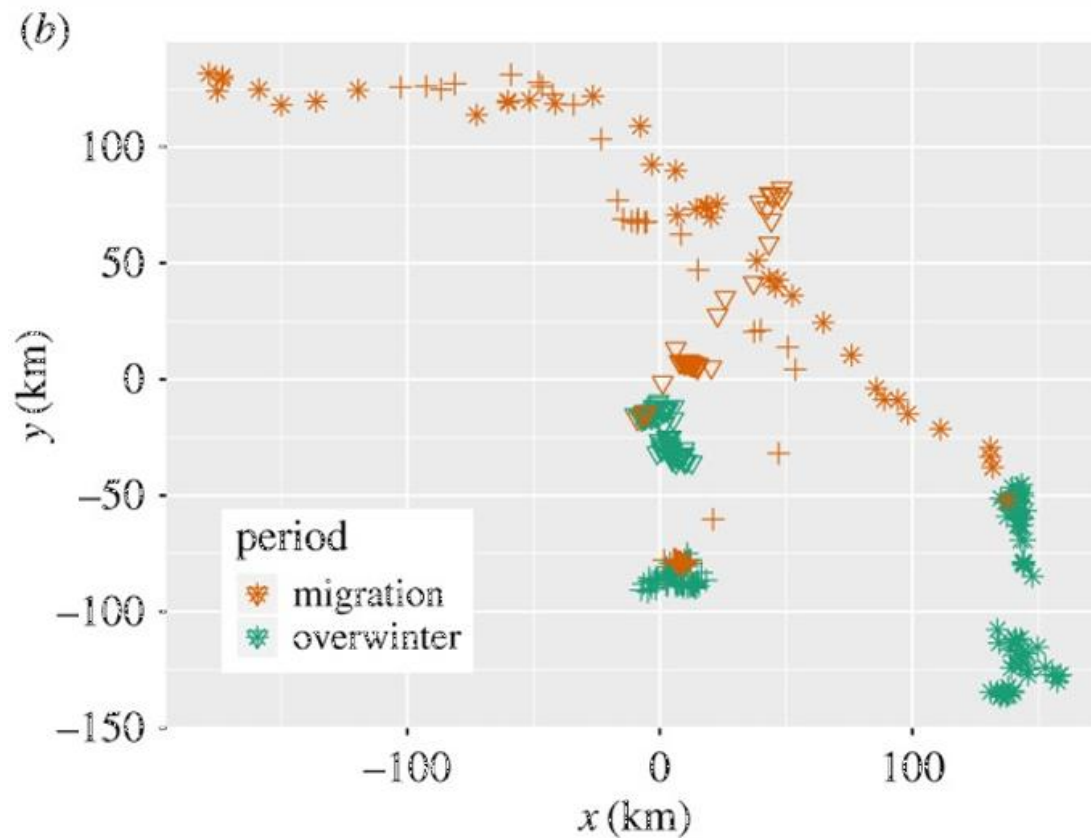
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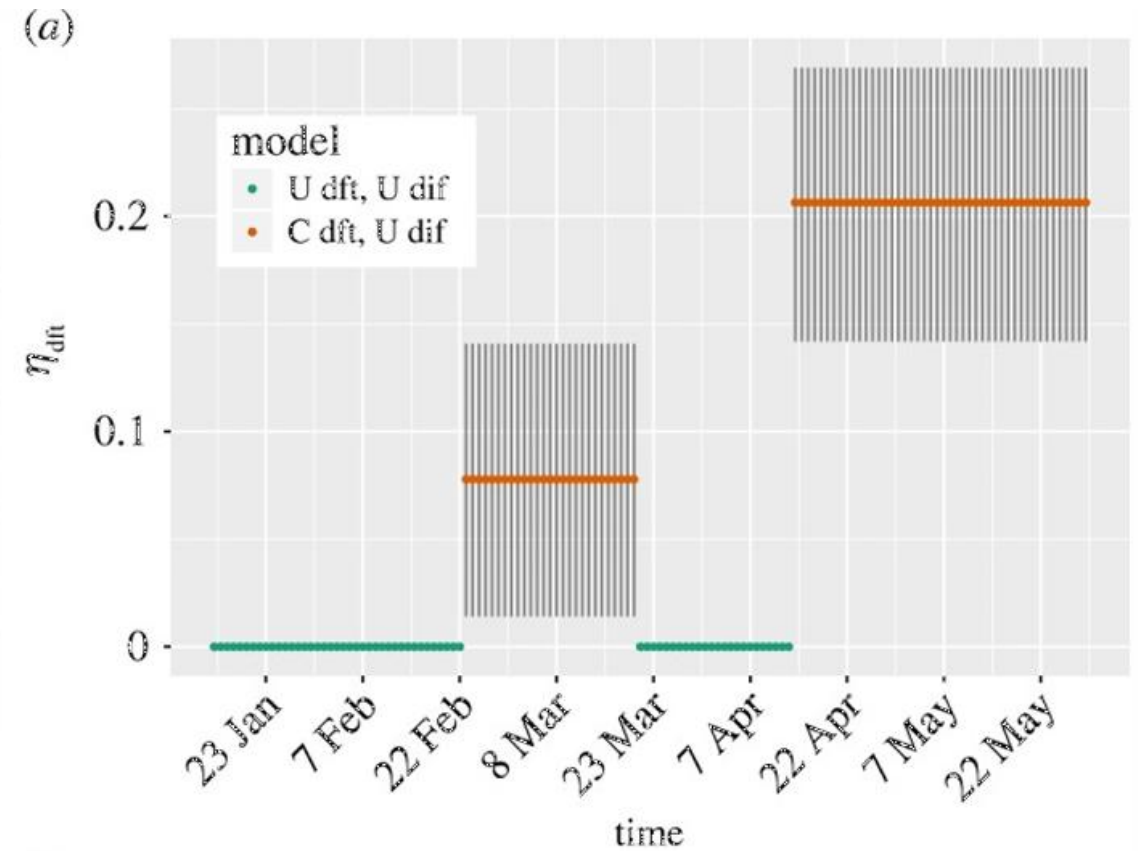
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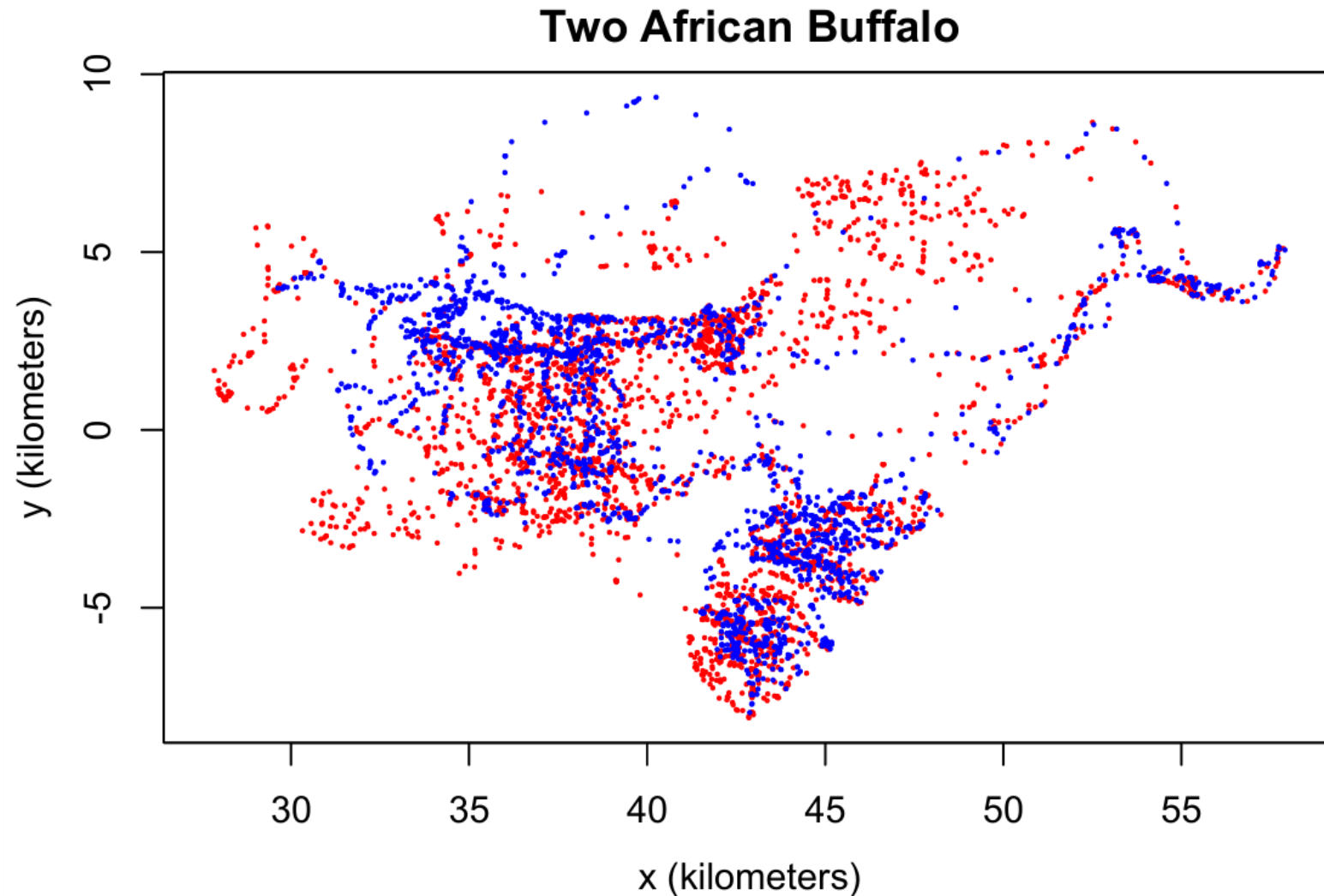
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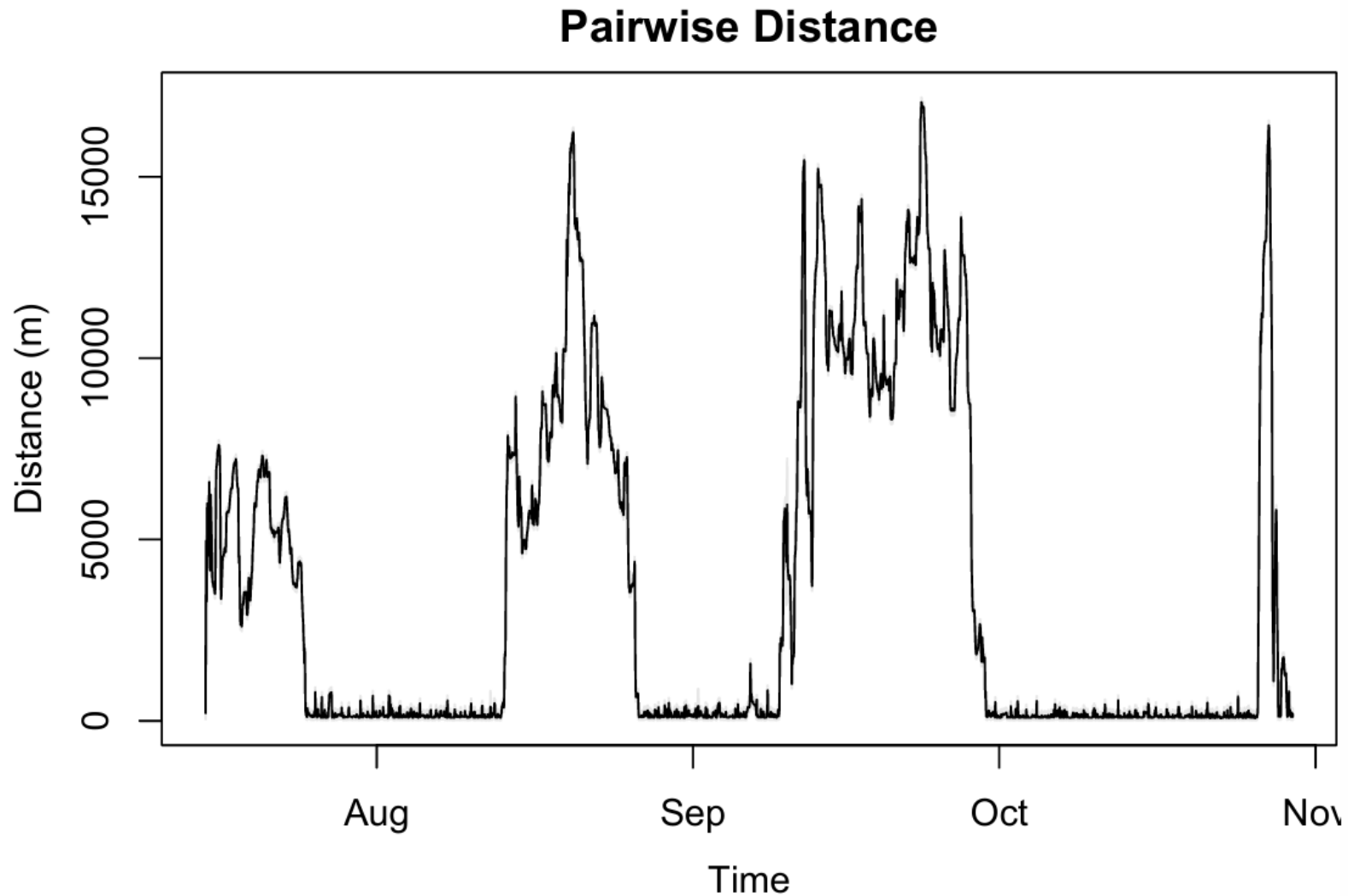
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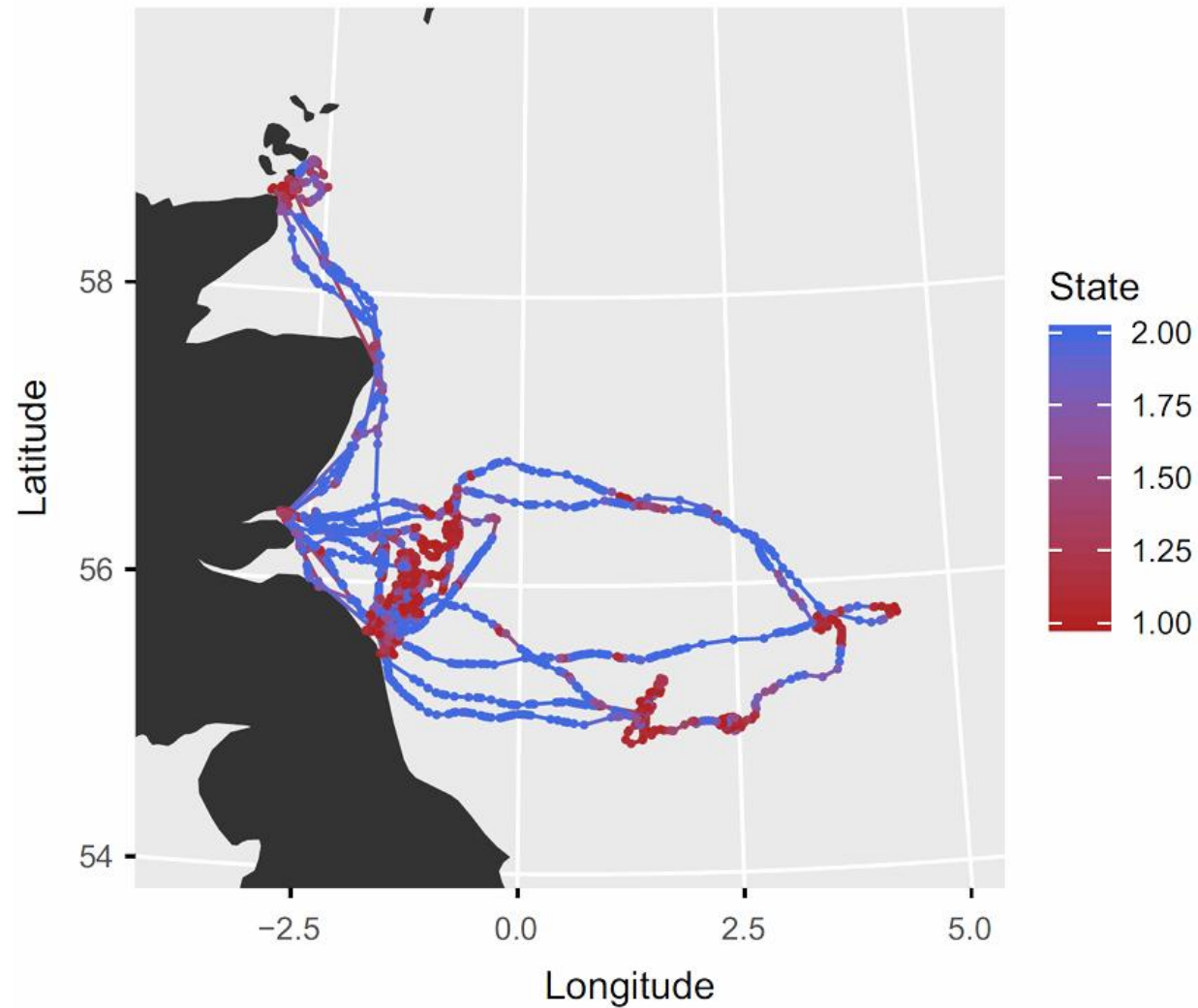
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Behavior switching model



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 - Sparse data, irregular sampling, location error, few observed encounters
- These challenges can be addressed with methods based on continuous-time stochastic process models
- Current methods are barely scratching the surface of what is possible
- Hopefully, we can address some of these challenges



ctmm_interactions.R

Thank you

- Thanks to collaborators: Jesse Alston, Justin Calabrese, Bill Fagan, Ricardo Martinez-Garcia, Michael Noonan, Dan Sheldon, Inês Silva, etc.
- Funded by and supported by:

